



MKS SERV0 42ES/57ES Closed-Loop Stepper Motor

RS485 User Manual V1.0.1

Note: This manual corresponds to firmware version V 1.0.1 .

MKS SERV042ES/57ES RS485 Version Notes			
manual	content	Firmware	date
V1.0.0	First release version	V1.0.0	Dec-2024
V1.0.1	1. Fixed some bugs	V1.0.1	Nov-2025
	2. Add IAP firmware upgrade function		
	3. F6 command speed mode adds the function of setting the runtime.		



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Part 1. Product Overview

1.1 Product Introduction

The MKS SERV042ES/57ES _RS485 closed-loop stepper motor is a product independently developed by the Maker Base to meet market demands and meet industrial-grade standards . It features a pulse interface and an RS485 interface, a built-in high-efficiency FOC vector algorithm , and a high-precision encoder . Through position feedback, it effectively prevents step loss. It is suitable for applications such as small robotic arms , 3D printers , engraving machines, writing machines , automation products , and e-sports competitions .

1.2 Features

1. Supports 6 operating modes : pulse interface (pulse-pulse, pulse-direction mode) , serial interface (open-loop, closed-loop mode);
2. High-performance FOC vector control algorithm, torque , speed, and position three-loop control ;
3. It supports curved acceleration and deceleration, resulting in smoother motor start-up and stopping;
4. Supports both relative and absolute position control modes;
5. Built-in 256-step subdivision interpolation algorithm , ultra- quiet motor operation , ultra-low vibration;
6. Maximum input pulse frequency : 300kHz; maximum speed: 3000 RPM.
7. Motor angle information is updated in real time (motor enabled or disabled) ;
8. Supports MODBUS -RTU communication protocol;
9. It has built-in stall protection function ;
10. Quick factory reset function;
11. High-speed performance is stable, operation is smooth and vibration-free, and emergency stop is possible ;
12. The integrated aluminum alloy casing provides effective heat dissipation and ensures more stable continuous high-current operation of the motor .
13. Provides an open-source host computer and STM32/ Arduion usage examples.
14. Supports left and right limit functions.
15. Onboard industrial-grade high-precision magnetic encoder ;
16. Onboard high - power MOSFET , 60V/56A;
17. It features an onboard RS-485 interface, 256 addresses , and supports broadcast and packet addresses .
18. Industrial-grade design and selection, stable and reliable;
19. Operating voltage: 20V~48V ;

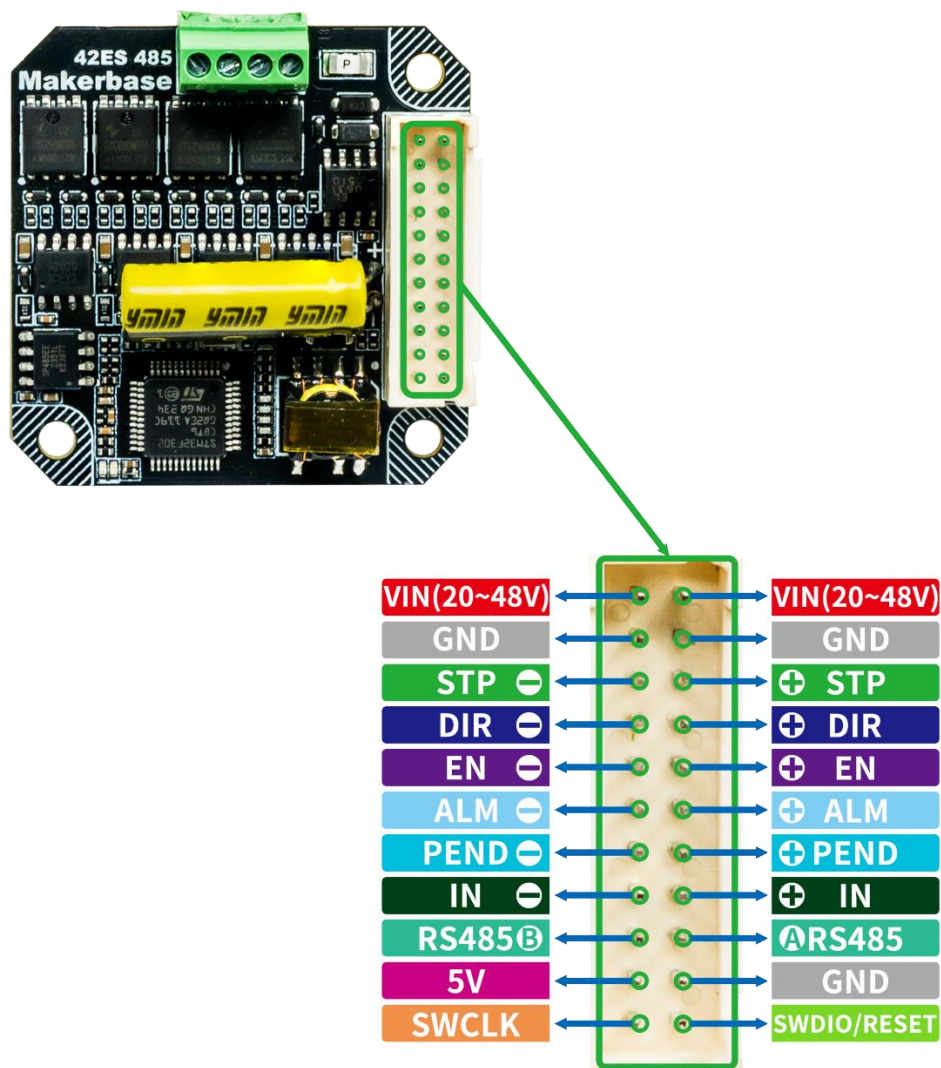


1.3 Product Parameters

Product Parameters		
Motherboard Model	MKS SERV042ES V1.0	MKS SERV057ES V1.0
MCU main controller	STM32F302CBT6	STM32F302RBT6
MOSFET	HYG090ND06LS1C2	
Magnetic encoder	MT6826	
RS485 transceiver	SP485EEN	
Operating voltage	20-48V	
Operating current	0-3000mA	0-5200mA
Closed-loop feedback frequency	Torque ring 20KHz	
	Speed loop 10kHz	
	Position ring 10KHz	
Maximum speed	3000RPM+	
Subdivision support	2/4/8/16/32/64/128/5/10/20/25/40/50/100/200/256	
Wiring method	Common anode, common cathode, differential, double pulse, pulse + direction	
Input signal level	3.3V-24V	
Pulse signal frequency	Up to 300kHz	
RS485 interface speed	9 600/19200/.../115200/25600	
RS485 interface address	1 broadcast address, 255 addresses	

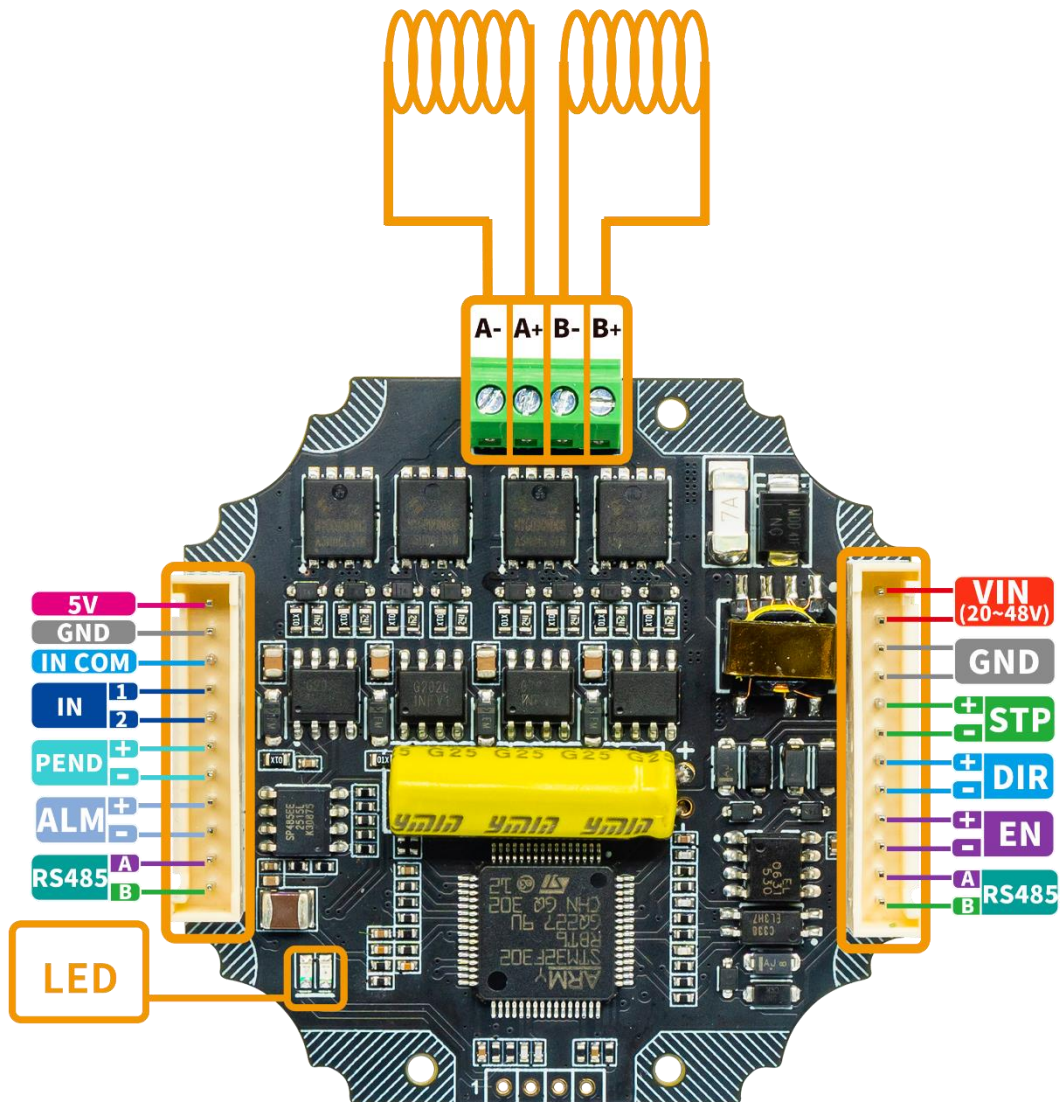
1.4 Interface Description

① SERVO42ES_ RS485 Interface Description



1. Input power supply 20V~48V
2. 5V is the output power (5V, 100mA).
3. ALM is an alarm signal (the optocoupler closes when an alarm occurs).
4. PEND is the positioning signal (after positioning, the optocoupler closes).
5. IN is the input signal, IN is the zero-return switch or left limit switch, EN can be remapped to the right limit switch. For details on remapping, please refer to the limit switch setting instructions in Part 4 (when EN is mapped to the right limit switch, the pulse control mode is disabled).

② SERV057ES RS485 Interface Description



1. Input power supply 20V~48V
2. 5V is the output power (5V, 100mA).
3. ALM is an alarm signal (the optocoupler closes when an alarm occurs).
4. PEND is the positioning signal (after positioning, the optocoupler closes).
5. IN COM is the common port of IN_1 and IN_2 (can be connected in either common cathode or common anode configuration).
6. IN_1 and IN_2 are input signals. IN_1 is the zero-return switch or left limit switch, and IN_2 is the right limit switch.



1.5 Port Function Description

Port Function Table			
PIN	I/O	Function	Description
STP+ (CW+)	Input	Pulse signal (or CW pulse) positive	(1) Optocoupler input, falling edge trigger. The pulse goes one step when the pulse goes from high to low. (2) Positive voltage: 3.3 - 28V. (3) Negative voltage: High level 3.3 - 28V, Low level 0 - 0.5V. (4) Maximum input frequency 400kHz, pulse duration greater than 2.5us. (5) Single pulse (Pulse + Dir) or double pulse (CW + CCW) is set by the command.
STP- (CW-)		Pulse signal (or CW pulse) negative	
DIR+ (CCW+)		Direction signal (or CCW pulse) positive	
DIR- (CCW-)		Direction signal (or CCW pulse) negative	
EN+		Enable signal positive	(1) Optocoupler input, low level effective. When ineffective, the motor will be released while clearing the alarm signal. (2) Positive voltage: 3.3 - 28V. (3) Negative voltage: High level 3.3 - 28V, Low level 0 - 0.5V.
EN-		Enable signal negative	
ALM+	Output	Alarm signal positive	(1) When an alarm occurs, the optocoupler output is on. (2) Maximum voltage +35V, maximum current 50mA.
ALM-		Alarm signal negative	
PEND+		In Position signal positive	(1) When the drive has finished a given pulse, the optocoupler output is on. (2) Maximum voltage +35V, maximum current 50mA.
PEND-		In Position signal negative	
A-	Output	Motor phase A-	Connected to motor phase line Note: If the phase sequence is not connected accordingly, the motor will alarm.
A+		Motor phase A+	
B-		Motor phase B-	
B+		Motor phase B+	
VDC	Input	Drive power supply positive	Operating Voltage +20 - 48V
GND		Drive power supply negative	



1.6 Indicator Light Status Description Table

LED Status	
Green stays on	Motor Running
Green flash	Motor Stop
1 Green + 1 Red	Over Current
1 Green + 2 Red	Open-Phase
1 Green + 3 Red	Supply Voltage High
1 Green + 4 Red	Supply Voltage Low
1 Green + 5 Red	Position Error
1 Green + 6 Red	Encoder Error

Tip: Please make sure the motor phase line is connected correctly, otherwise there will be a tracking error alarm after receiving the pulse (5 red and 1 green).

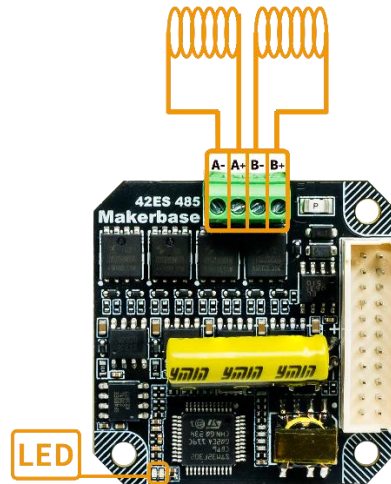
Part 2. Wiring method

2.1 Motor wiring method

Note: The internal resistance of the motor should be less than 10 ohms.

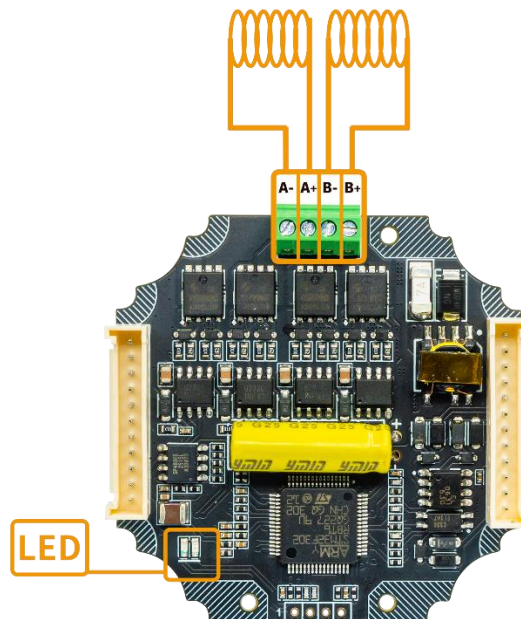
A+ and A- to one phase wire of the motor, and connect B+ and B- to the other phase wire of the motor.

① SERV042ES Wiring Method



Note: Please ensure the motor phase wires are connected correctly; otherwise, a tracking error exceeding tolerance alarm (5 red, 1 green) will occur upon receiving the pulse.

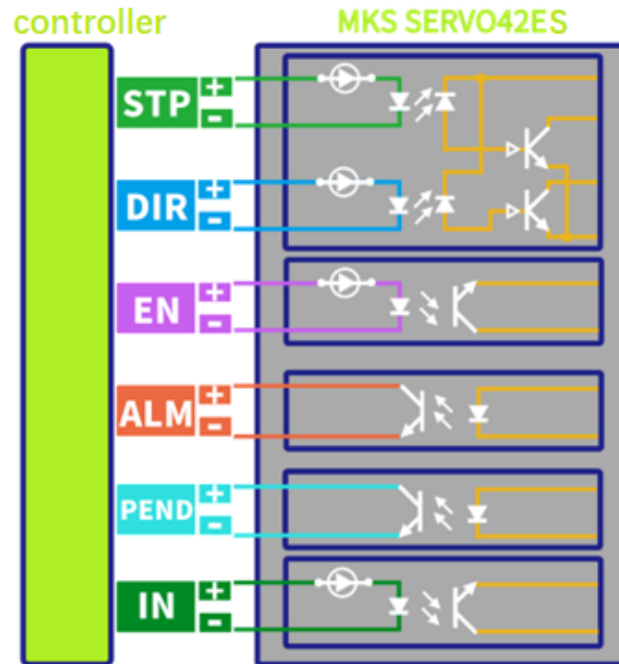
② SERV057ES Wiring Method



Note: Please ensure the motor phase wires are connected correctly; otherwise, a tracking error exceeding tolerance alarm (5 red, 1 green) will occur upon receiving the pulse.

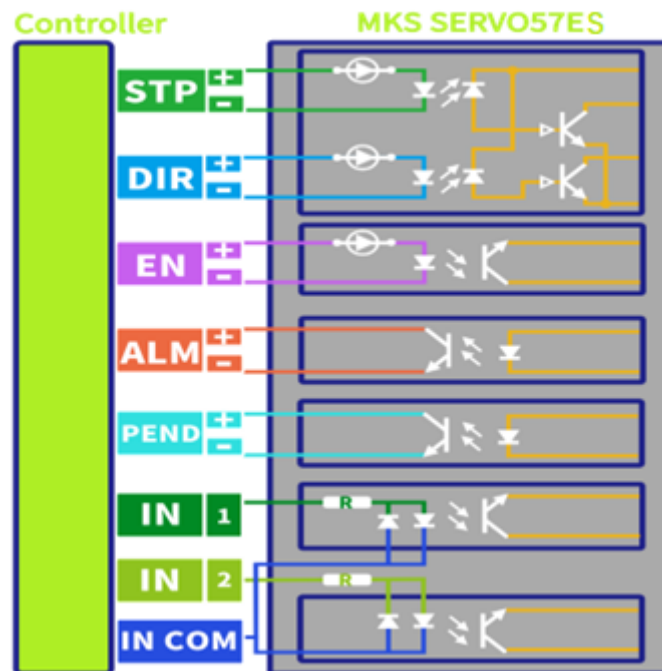
2.2 Pulse control wiring method

① SERV042ES Wiring



Note: All input ports have a built-in 10mA current limiter, allowing direct input of 3.3V-24V signals without the need for external current-limiting resistors.

② SERV057ES Wiring



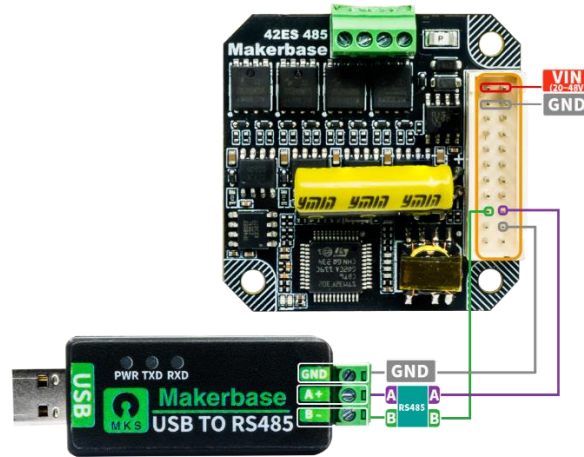
Note: All input ports have a built-in 10mA current limiter, allowing direct input of 3.3V-24V signals without the need for external current-limiting resistors.

2.3 RS485 Wiring Method

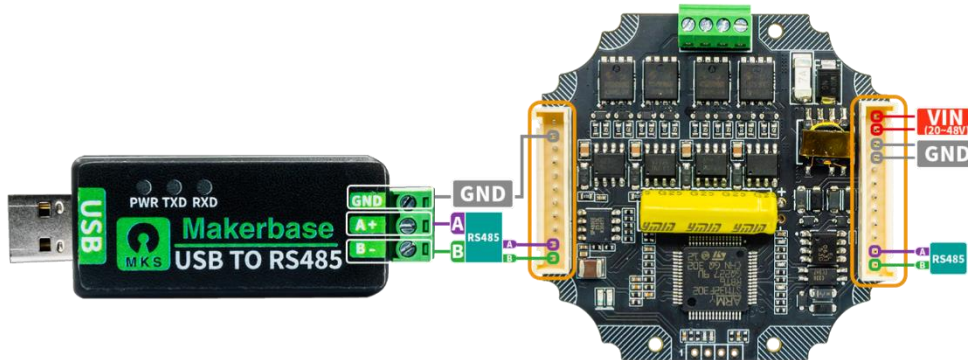
1. RS485 standalone communication wiring

Note: To reduce bus interference, the host computer and the motor must share a common ground. RS485 A and B signals should be transmitted using shielded twisted-pair cable.

① MKS SERVO42ES RS485 standalone wiring

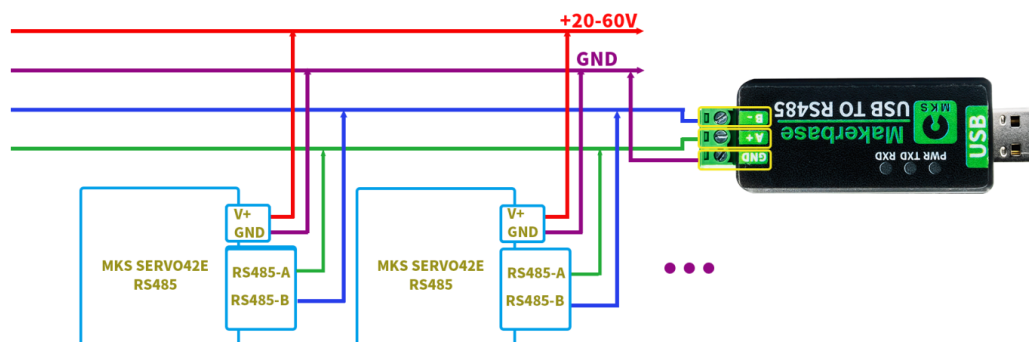


② MKS SERVO57ES RS485 standalone wiring



Note: The SERVO57ES board has two RS485 interfaces connected in parallel; connecting to either one is sufficient.

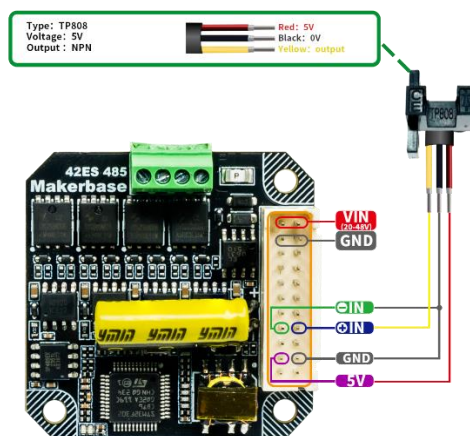
2. RS485 multi-machine communication wiring



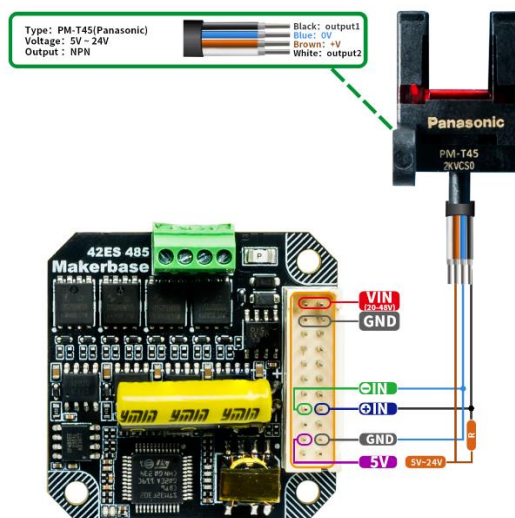
2.4 External switch wiring method

① MKS SERVO42ES Switch Wiring Method

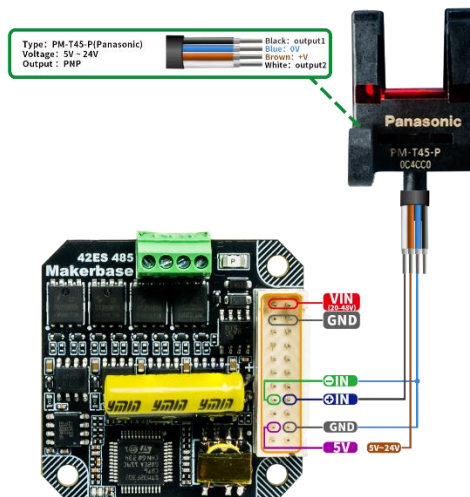
a) TP808 Wiring Method



b) PM-T45 Wiring Method

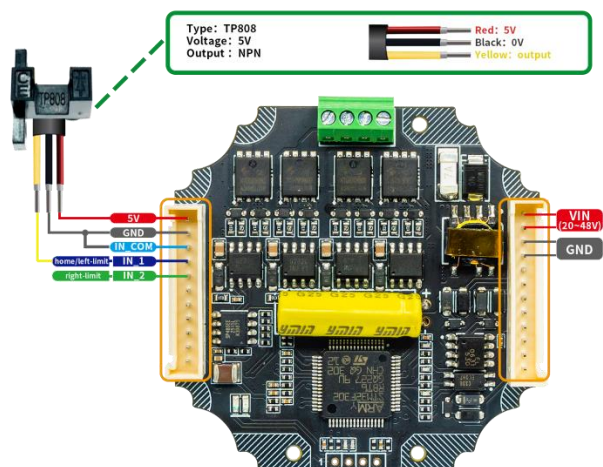


c) PM-T45-P Wiring Method

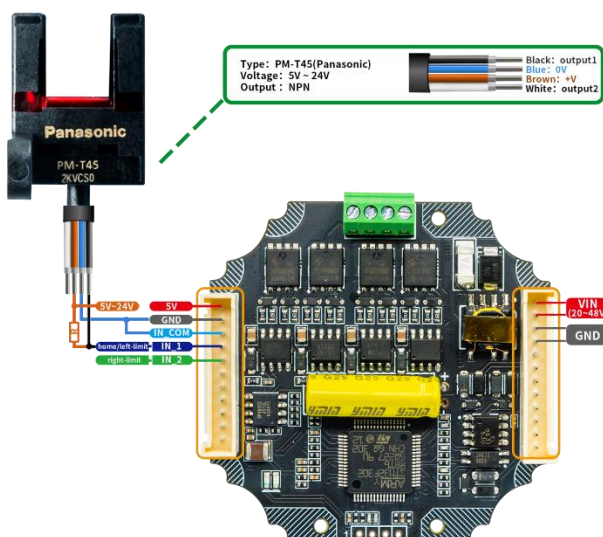


② MKS SERVO57ES Switch wiring method

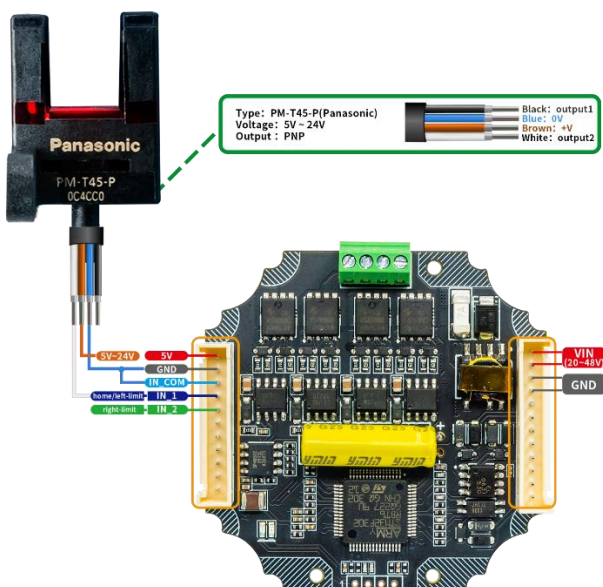
a) TP808 Wiring Method



b) PM-T45 Wiring Method



c) PM-T45-P Wiring Method



Part 3. Serial data format description

Downlink frames (PC → SERVOxxES)							
Header	Address	Code	Instruction data				CRC check code
FA	addr	code					CRC
Uplink frames (PC ← SERVOxxES)							
Header	Address	Code	Returned data				CRC check code
FB	addr	code					CRC

1. Instruction data and return data are stored in big-endian format.
2. Downlink header FA , uplink header FB.
3. The address (addr) ranges from 0 to 255 , with the default address being 01.

Where 00 is the broadcast address ;

4. The code executes the corresponding instruction ;
for example, 0x80 executes the calibration instruction .
5. For instructions or return data , please refer to the "Serial Port Instruction Description ".
6. CRC checksum is CHECKSUM 8bit

For example, the instruction "FA 01 80 00 CRC"

$$\text{CRC} = (0\text{xFA} + 0\text{x01} + 0\text{x80} + 0\text{x00}) \& 0\text{xFF} = 0\text{x17B} \& 0\text{xFF} = 0\text{x7B}$$

7. When the host computer sends instructions, the timing between bytes of a single instruction (FA ... CRC) must be continuous, and there cannot be a delay of more than one byte; otherwise, the slave computer may fail to receive the instruction.

Note: When sending commands using broadcast addresses or packet addresses, the slave device will not respond.



Part 4. General Instructions

Note 1: When sending commands using broadcast addresses or packet addresses, the slave device will not respond.

Note 2: All write-only instructions are not saved immediately after being set; the 60H instruction must be executed to save them all at once.

4.1 Calibrate motor instructions

Note: The complete machines sold by the manufacturer are already calibrated at the factory and do not require recalibration.

Calibration can compensate for and correct nonlinearities caused by magnet installation deviations.

If the motor phase wires are wired according to the factory default connection, no calibration is required. Although it can be used, the motor performance may be poor.

To ensure the motor performs optimally, or if issues such as excessive motor vibration or inconsistent current (sound) in both forward and reverse directions occur, the motor can be recalibrated.

The motor must be unloaded before calibration and must not be under any load.

During calibration, the motor rotates at a speed of 600 RPM for more than 20 revolutions.

If calibration is successful, the green light will flash rapidly. Power off and restart the motor before it can be used normally.

If calibration fails, the red light will flash rapidly. You need to power off and restart the motor, and then calibrate again.

Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Data	Checksum
FA	01 H	80H	01H	CRC

No data was returned.

After receiving the command, the motor will automatically restart and enter calibration mode. You need to wait about 10 seconds to observe the status of the LED lights and judge the calibration result.



4.2 Read-only parameter instructions

4.2.1 Reading the value of a multi-turn encoder

It can read the motor rotation range recorded by the encoder after power-on.

Note: Encoder single-turn value range is 0~0x4000

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	30H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)					
Byte1	Byte2	Byte3	Bytes 4-7	Bytes 8-9	Byte10
Header	Address	Code	Number of laps	Encoder value	Checksum
FB	addr	30H	carry(int32_t)	value	CRC

carry: the carry vaule of the encoder.

value: the current vaule of the encoder.(range 0~0x3FFF)

When value is greater than 0x3FFF, carry +=1.

When Value is less than 0, carry -=1.

For example:

If the current carry|value is 0x3FF0, After one turn CCW,the carry|value (+0x4000) is 0x13FF0.

If the current carry|value is 0x3FF0, After one turn CW,the carry|value (-0x4000)is 0xFFFFFFF3FF0.

Note: The encoder value is updated regardless of whether the motor is enabled or not.



4.2.2 Read the cumulative multi-turn encoder value

It can read the motor rotation range recorded by the encoder after power-on.

Note: Encoder single-turn value range is 0~0x4000

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	31H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-9	Byte10
Header	Address	Code	Encoder value	Checksum
FB	addr	31H	value (int48_t)	CRC

value: Records the multi-turn encoder value, data type int48_t

After one turn clockwise,the value += 0x4000;

After one turn CCW,the value -= 0x4000;

For example:

If the current value is 0x3FF0, After one turn CCW,the value(+0x4000) is 0x7FF0.

If the current value is 0x3FF0, After one turn CW,the value(-0x4000) is 0xFFFFFFFFFF0.

Note: When moving relative to or absolutely according to coordinate values, use the encoder value as the coordinate.

4.2.3 Read the real-time speed of the motor

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	32H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-5	Byte6
Header	Address	Code	motor speed	Checksum
FB	addr	32H	rpm	CRC

rpm: Real-time motor speed, unit RPM, data type int16_t

Note: The counterclockwise rotation speed is greater than 0, and the clockwise rotation speed is less than 0.



4.2.4 Read the number of received pulses

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	33H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-7	Byte8
Header	Address	Code	pulse count	Checksum
FB	addr	33H	pulses	CRC

pulses: The number of pulses received, data type int32_t

4.2.5 Read motor alarm status

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	37H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Motor status	Checksum
FB	addr	37H	status	CRC

status: Motor status, corresponding codes are shown in the table below.

Motor status	status	Remark
motor running	0	
Motor stopped	1	
Over current error	2	
Phase loss error	3	
Overvoltage error	4	
Undervoltage error	5	
Position error	6	
Encoder error	7	



4.2.6 Read position angle error

Angle error is the difference between the position angle controlled by the command and the real-time angle position of the motor. The unit is $0 \sim 51200$, which means $0 \sim 360^\circ$. For example, when the error is 1° , the value is $51200 / 360^\circ = 142.22$, and so on.

The read command is as follows:

Downlink frames (PC $\rightarrow$ SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	39H	CRC

Returned data:

Uplink frames (PC $\leftarrow$ SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-7	Byte8
Header	Address	Code	Error value	Checksum
FB	addr	39H	errors	CRC

errors: Real-time position error value, data type int32_t

4.2.7 Read motor enable status

When using bus control, this command can be used to obtain the enable status of the driver board.

The read command is as follows:

Downlink frames (PC $\rightarrow$ SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	3AH	CRC

Returned data:

Uplink frames (PC $\leftarrow$ SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Enable state	Checksum
FB	addr	3AH	status	CRC

status=01 Motor is enabled

status=00 Motor not enabled



4.2.8 Read the motor return to zero status

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	3BH	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Return to zero state	Checksum
FB	addr	3BH	status	CRC

status=00 The motor has not returned to zero, or the return-to-zero process failed.

status=01 The motor has returned to zero.



4.2.9 Automatically return read-only parameters at regular intervals

This command can set parameters so that the motor can automatically return a certain read-only parameter at regular intervals, without the host computer needing to frequently issue corresponding query commands.

The command format is as follows:

Downlink frames (PC → SERV0xxES)					
Byte1	Byte2	Byte3	Byte4	Bytes 5-6	Byte7
Header	Address	Cmd	Code	Timer	Checksum
FA	addr	01H	code	times	CRC

code: The code corresponding to the read-only parameter.

times: The timeout period, in milliseconds (when times = 0, no data is returned).

Returned data:

Uplink frames (PC ← SERV0xxES)					
Byte1	Byte2	Byte3	Byte4	Byte5	Byte6
Header	Address	Cmd	Code	state	Checksum
FB	addr	01H	code	status	CRC

status=00 Setup failed

status=01 Successfully set

The parameter data format returned periodically is as follows:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte 4-n	Byte n+1
Header	Address	Code	Parameter value	Checksum
FB	addr	code	param	CRC

param: The corresponding read-only parameter

Taking the reading of the "accumulator multi-turn encoder value" every 1000ms as an example

Send FA 01 01 31 03 E8 18

Return to FB 01 01 31 01 2F

After successful setup, the motor will return the "cumulative multi-turn encoder value" every 1000 milliseconds, as shown in the figure below.

```
[2025-10-11 17:18:30.020]# SEND HEX>
FA 01 01 31 03 E8 18
[2025-10-11 17:18:30.092]# RECV HEX>
FB 01 01 31 01 2F
[2025-10-11 17:18:31.081]# RECV HEX>
FB 01 31 00 00 00 00 04 31
[2025-10-11 17:18:32.098]# RECV HEX>
FB 01 31 00 00 00 00 04 31
[2025-10-11 17:18:33.089]# RECV HEX>
FB 01 31 00 00 00 00 04 31
[2025-10-11 17:18:34.085]# RECV HEX>
FB 01 31 00 00 00 00 04 31
```



4.2.10 Read a single configuration parameter

After writing the configuration parameters, you can use this command to read and view them.

The command format for reading write-only configuration parameters is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Cmd	Code	Checksum
FA	addr	00H	code	CRC

code: The code corresponding to the parameter-only function.

For example, to read the "working mode", the corresponding code is 82H.

The returned parameter data format is as follows:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte 4-n	Byte n+1
Header	Address	Code	Parameter value	Checksum
FB	addr	code	param	CRC

param: The corresponding configuration parameter

Note: The data format of param should be consistent with the data format when setting this parameter.

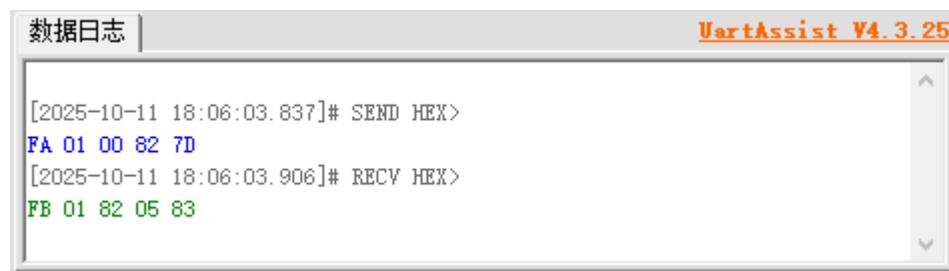
If this parameter does not support reading, the returned data will be as follows:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-5	Byte6
Header	Address	Code	Parameter value	Checksum
FB	addr	code	0xFFFF	CRC

Taking reading the "working mode" as an example:

Send FA 01 00 82 7D

Return to FB 01 82 05 83, as shown in the image below.





4.2.11 Read version information

Read hardware and firmware information.

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	40H	CRC

The returned data format is as follows:

Uplink frames (PC ← SERV0xxES)							
Byte1	Byte2	Byte3	Byte4			Bytes 5-7	Byte8
Header	Address	Code	b7	b5-b4	b3-b0	Firmware version	check
			D/E series distinction	Calibration status	Hardware version		
FB	addr	40	series	cal	hardVer	firmVer[3]	CRC

series = 1 E series stepper motor

series = 0 D series stepper motor

cal = 1 encoder value decreases as the motor rotates CW.

cal = 2 encoder value increases as the motor rotates CW.

Firmware version: firmVer[0] = 1, firmVer[1] = 0, firmVer[2] = 0

Corresponding version V1.0.0

The hardware version correspondence is as follows:

Board type	hardVer
S42ES_RS485	1
S42ES_BUS	2
S57ES_RS485	3
S57ES_BUS	4
S28ES_RS485	5
S28ES_BUS	6
S35ES_RS485	7
S35ES_BUS	8



4.2.12 Read /Write User-defined ID

1. Write the user ID number.

The command to write the ID is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-7	Byte8
Header	Address	Code	ID	Checksum
FA	addr	42H	ID	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Enable state	Checksum
FB	addr	42H	status	CRC

status=00 Write failed

status=01 Write successful

2. Read user ID

The command to read the ID is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	42H	CRC

Return ID data:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-7	Byte8
Header	Address	Code	ID	Checksum
FB	addr	42H	ID	CRC



4.3 Write-only parameter command

Note: The settings are not saved immediately after writing the instructions; you need to execute the 60H instruction to save them all at once.

4.3.1 Set working mode

The instructions for setting the working mode are as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Work mode	Checksum
FA	addr	82H	mode	CRC

The corresponding working modes are shown in the table below.

List of working modes		
	mode	Work mode
With encoder	00H	Pulse + Pulse Open-Loop FOC Mode
	01H	Pulse + Direction Open-Loop FOC Mode
	02H	Pulse + Pulse Closed-Loop FOC Mode
	03H	Pulse + Direction Closed-Loop FOC Mode (Default)
	04H	RS485 bus open-loop FOC mode
	05H	RS485 bus closed-loop FOC mode
No encoder	10H	Pulse + Pulse Open-Loop FOC Mode
	11H	Pulse + Direction Open-Loop FOC Mode
	14H	RS485 bus open-loop FOC mode

Note 1: An encoder is present, meaning the motor shaft has magnets. The drive board is mounted on the back and can read the encoder values.

No encoder, meaning the motor shaft has no magnets, the drive board can be installed arbitrarily, and the encoder value cannot be read.

Note 2: Pulse control mode, maximum input frequency 300kHz

Bus control mode, maximum speed 3000 RPM

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	82H	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.



4.3.2 Set operating current

The operating current setting instructions are as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-5	Byte6
Header	Address	Code	current (mA)	Checksum
FA	addr	83H	current	CRC

SERV042ES maximum operating current is 3000mA (default 1600mA).

SERV057ES maximum operating current is 5200mA (default 3200mA).

for example:

Send FA 01 83 02 30 B0 to set the operating current to 560mA.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	83H	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.

4.3.3 Set subdivisions

The detailed settings instructions are as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Value	Checksum
FA	addr	84H	MS	CRC

The MS value range (decimal) is as follows: (default value 16)

0, 2, 4, 8, 16, 32, 64, 128,

5, 10, 20, 25, 40, 50, 100, 200

Note: 0 corresponds to 256 subdivisions.

for example:

Send FA 01 84 10 8F, set to 16 subdivisions.

Send FA 01 84 0A 89, set to 10 subdivisions.

Send FA 01 84 00 7F, set to 256 subdivisions.

And so on...

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	84H	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.



4.3.4 Set the active level of the En pin.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	effective level	Checksum
FA	addr	85H	Level	CRC

Level = 00 corresponds to low-level enable (L) (default).

Level = 01 corresponds to high-level enable (H).

Level = 02 corresponds to always being enabled (Hold).

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	85H	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.

Note 1: After successful setup, pulse signals will be received after a 100ms delay.

Note 2: Only valid for pulse control mode.

4.3.5 Set the motor rotation direction

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	direction	Checksum
FA	addr	86H	dir	CRC

dir = 00 Corresponds to clockwise rotation (default)

dir = 01 Corresponding to counterclockwise rotation

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	86H	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.

Note: This command can also change the direction of motor operation controlled by the bus.



4.3.6 Set pulse delay

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Pulse delay	Checksum
FA	addr	87H	delay	CRC

delay = 00 delay 0ms
delay = 01 delay 4ms
delay = 02 delay 20ms (default)
delay = 03 delay 40ms

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	87H	status (uint8_t)	CRC

status = 0 set failed.
status = 1 set succeeded.

4.3.7 Set the serial port baud rate

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	baud rate	Checksum
FA	addr	8AH	baudrate	CRC

baudrate = 01 corresponds to a baud rate of 9600.
baudrate = 02 corresponds to a baud rate of 19200.
baudrate = 03 corresponds to a baud rate of 25000.
baudrate = 04 corresponds to a baud rate of 38400 (default).
baudrate = 05 corresponds to a baud rate of 57600.
baudrate = 06 corresponds to a baud rate of 115200.
baudrate = 07 corresponds to a baud rate of 256000.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	8AH	status (uint8_t)	CRC

status = 0 set failed.
status = 1 set succeeded.



4.3.8 Set address

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Address	Checksum
FA	addr	8BH	address	CRC

address is the address, ranging from 00 to 0xFF.

Note : 00 is the broadcast address, 01 is the default address.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	8BH	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.



4.3.9 Configure slave response method

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Response method	Checksum
FA	addr	8CH	ackMode	CRC

ackMode = 00, no response from the slave device.

ackMode = 01, the slave device passively acknowledges, meaning it does not actively initiate data transmission.

ackMode = 02, Slave device actively responds (default value)

Example of different response methods, taking position control mode 1 as an example:

The host sends FA 01 FD 02 80 02 00 00 FA 00 76

a. No response mode (ackMode = 00)

The slave device does not return any information.

b. In passive response mode (ackMode = 01)

Slave immediately returns to position control start 01 or fails 00

c. In active response mode (ackMode = 02)

Slave immediately returns to position control start 01 or fails 00

After the motor finishes running or stops when it hits the limit switch, return to 02 or 03.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	8CH	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.

Note: After setting the slave device to no response, the motor operating status can be queried using the code "F1H".



4.3.10 Configure MODBUS -RTU communication protocol

Enabling this function will use the MODBUS-RTU communication protocol instead of the MKS communication protocol defined in this manual.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	RTU Protocol	Checksum
FA	addr	8EH	enable	CRC

enable = 00, Disable MODBUS-RTU protocol (default).

enable = 01, Enable MODBUS -RTU protocol

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	8EH	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.

Note: In MODBUS-RTU communication protocol mode, if you need to switch back to using the MKS communication protocol, you can send the following command to disable the MODBUS-RTU communication protocol.

“ 01 06 00 18 00 00 09 CD” (Setting up the MKS protocol)

“ 01 06 00 54 00 01 09 DA” (Save parameters)

“ 01 06 00 56 00 01 A8 1A” (Reset and restart the motor)

After a reset and restart, the system will enter the MKS protocol.

4.3.11 Configure whether to lock the axis when starting in bus mode.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Locking shaft	Checksum
FA	addr	8FH	enable	CRC

enable = 00, the shaft is not locked when starting in bus mode.

enable = 01, the shaft is locked when starting in bus mode. (default)

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	8FH	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.



4.3.12 Set group address

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Group address	Checksum
FA	addr	8DH	groupAddr	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	8DH	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.

Application instructions for using packet addresses:

Assuming there are 6 motors, the address settings are as follows:

	Broadcast address	Address	Group address
Motor 1	0	1	0x 50
Motor 2	0	2	0x 50
Motor 3	0	3	0x 50
Motor 4	0	4	0x 51
Motor 5	0	5	0x 51
Motor 6	0	6	0x 51

Send FA 01 FD 01 2C 64 00 00 0C 80 15 Motor 1 Rotate 1 full circle

Send FA 00 FD 01 2C 64 00 00 0C 80 14 Motor 1-6 Rotate 1 full circle

Send FA 50 FD 01 2C 64 00 00 0C 80 64 Motor 1-3 Rotate 1 full circle

Send FA 51 FD 01 2C 64 00 00 0C 80 65 Motor 4-6 Rotate 1 full circle

Note: When sending commands using a packet address, the slave device will not respond.



4.3.13 Set heartbeat protection time

Heartbeat protection refers to the system that, if the motor does not receive any instructions from the host computer within the set protection time, it will control the motor to stop urgently, preventing abnormal accidents caused by communication interruption.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Bytes 4-7	Byte8
Header	Address	Code	Protection time	Checksum
FA	addr	89H	times	CRC

times: Protection time, in milliseconds. (Default: 0)

Note: When times=0, the heartbeat protection function is turned off.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	89H	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.

4.3.14 Set position to reach threshold

In bus position control mode, the "position operation completed" message will only be returned when the actual position of the motor is less than the threshold.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)					
Byte1	Byte2	Byte3	Byte4	Bytes 5-6	Byte7
Header	Address	Code	Enable	Reaching the threshold	Checksum
FA	addr	98H	enable	values(uint16_t)	CRC

enable: 00 Disable position reaches threshold judgment

 01 Enable location arrival threshold judgment (default)

values: Position threshold (default value 800)

Note: The maximum value is 65535, which is 360 degrees.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	98H	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.



4.4 Stall protection instructions

If the motor fails to reach the designated position within the response time due to overload or any other reason, the stall protection will be triggered, the motor will unlock and the shaft will be released to avoid damage.

4.4.1 Set stall protection and out-of-tolerance settings.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)					
Byte1	Byte2	Byte3	Byte4	Bytes 5-6	Byte7
Header	Address	Code	Stall protection	Excess tolerance	Checksum
FA	addr	88H	enable	value	CRC

enable = 00 Disables stall protection.

enable = 01 Enables stall protection (default)

value: Range of out-of-tolerance values (0-0x7FFF), default value 0x64

When the error exceeds the value, the stall protection is triggered, and the motor shaft is released.

The value = 0x64 corresponds to an angle of 180 degrees.

The value = 0xC8 corresponds to an angle of 360 degrees.

And so on.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	88H	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.

After stall protection

- 1.the stall state can be released by sending a command (3DH).
- 2.the stall state can be released by loosening the motor shaft.



4.4.2 Read motor stall status

When a motor stalls, a stall flag will be set. This command can be used to determine whether a stall has occurred. If the stall protection option is enabled, the drive board will automatically shut down the driver upon a stall.

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	3EH	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Enable state	Checksum
FB	addr	3EH	status	CRC

status=00 Motor not stalled

status=01 Motor is stalled.

4.4.3 Release the motor from stall state

When the motor stalls, sending this command can release the current stall state.

If a stall occurs again after the stall is cleared, the stall protection will still be triggered.

The release command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	3DH	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Enable state	Checksum
FB	addr	3DH	status	CRC

status=00 Release failed

status=01 Release successful

Note: the stall state also can be released by loosening the motor shaft.



4.5 Save /Restore Parameters and Reset Instructions

4.5.1 Save parameter command

This command saves all modified parameters to internal storage.

The command to save parameters is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	control word	Checksum
FA	addr	60H	01H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Save status	Checksum
FB	addr	60H	status (uint8_t)	CRC

status = 0 Save failed

status = 1 Save succeeded

4.5.2 Restore factory settings

The command to restore factory settings is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	3FH	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Restored status	Checksum
FB	addr	3FH	status (uint8_t)	CRC

status = 0 Recovery failed

status = 1 Recovery successful

Note : The driver board will automatically restart after restoring factory settings.



4.5.3 Reset and restart the motor

The command to restore factory settings is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	41H	CRC

Note: This command only resets and restarts the motor; it does not modify configuration parameters.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	41H	status (uint8_t)	CRC

status = 0 Reset failed

status = 1 Reset successful



4.6 IAP Firmware Online Upgrade Instructions

IAP firmware upgrades do not require disassembling or recalibrating the motor, the user configuration parameters will be retained after firmware upgrade.

Instructions for IAP upgrades can be found in "MKS SERVO42&57E IAP Upgrade Instructions.pdf".

The IAP upgrade operation video can be found in "MKS SERVO42&57E IAP Upgrade Operation Video.mp4".

IAP Mode are as follows:

Downlink frames (PC → SERVOxxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	control word	Checksum
FA	addr	50H	cmd	CRC

cmd = 01 Enter boot mode

cmd = 02 Enter silent state

cmd = 03 Exit silent state

Control word cmd description:

When there is only one motor on the bus, send the cmd = 01 command directly to enter boot mode;

When there are multiple motors on the bus, the following steps can be taken to avoid data interference:

- First, broadcast the command cmd = 02 (FA 00 50 02 CRC) to put all motors into a silent state;
- Then send the command cmd = 01 to the motor to be upgraded to enter boot mode and upgrade the firmware;
- After the upgrade is complete, broadcast the command cmd = 03 (FA 00 50 03 CRC) to make all motors exit the silent state.

Note: In silent mode, the motor does not respond to commands other than 50H.

Note: After cmd = 01 is successfully executed, the motor will automatically restart and enter IAP mode 2, waiting to receive the upgrade file.

Returned data:

Uplink frames (PC ← SERVOxxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	50H	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.



4.7 Complete parameter description for a single read/write operation

4.7.1 Command to write all configuration parameters

The write command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4-Byte52	Byte53
Header	Address	Code	Parameter value	Checksum
FA	addr	46H	params	CRC

The values for the parmas parameter are defined in the table below.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	46H	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.

The default write parameters are as follows:

```
FA 01 46 03 FF 0C 80 10 00 00 02 01 00 64 FF 00 00 00 00 04 01 02
00 00 01 00 00 00 64 00 0A 02 58 00 00 20 00 FF 00 00 00 EA 60 00
00 00 00 00 00 00 00 00 7E
```

(The last byte 7E is the checksum)

4.7.2 Command to read all configuration parameters

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	47H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4-Byte52	Byte53
Header	Address	Code	Parameter value	Checksum
FB	addr	47H	params	CRC

The values for the parmas parameter are defined in the table below.



Write all parameters params			
byte	format	instruction	Default (HEX)
1	FA		
2	1		
3	46H		
4	Work mode	82	3
5	reserve		FF
6	Operating current	83	0C80 / 0640
7			
8	micStep	84	10
9	En level	85	0
10	direction	86	0
11	Pulse delay	87	2
12	Stall protection	88	1
13	out-of-		64
14	olerance values		
15	reserve		FF
16	Heart rate protection time	89	0
17			
18			
19			
20	baud rate	8A	4
21	Address	8B	1
22	Resp method	8C	2
23	Group address	8D	0
24	MODBUS	8E	0
25	Bus- shaft lock	8F	1
26	Zero mode	93	0
27	zero direction		0
28	"High speed"		64
29			
30	"Low speed"		A
31			
32	Zero- current	94	0258/012C
33			Origin offset
34			
35			
36			
37			
38	reserve		FF

Read all parameters (params)	
byte	format
1	FB
2	1
3	47H
4	Work mode
5	reserve
6	Operating current
7	
8	micStep
9	En level
10	direction
11	Pulse delay
12	Stall protection
13	out-of- olerance values
14	
15	Reserved (FF)
16	Heart rate protection time
17	
18	
19	
20	baud rate
21	Address
22	Resp method
23	Group address
24	MODBUS
25	Bus- shaft lock
26	Zero mode
27	zero direction
28	"High speed"
29	
30	"Low speed"
31	
32	Zero- current
33	
34	Origin offset
35	
36	
37	
38	Reserved (FF)



Write all parameters command (46H)			
byte	Instruction format	Corresponding single instruction	Default value (hexadecimal)
39	trigger method	95	0
40	Zeroing timeout		EA60
41			
42			
43			
44	Limit enable	9E	0
45	Trigger level		0
46	endstop map		0
47	Start level	9F	0
48	Frequency division period		0
49			
50			
51			
52	Synchronous working mode	4A	0
53	Check value		

Read all parameters (params)	
byte	Instruction format
39	trigger method
40	Zeroing timeout
41	
42	
43	
44	Limit enable
45	Trigger level
46	endstop map
47	Start level
48	Frequency division period
49	
50	
51	
52	Synchronous working mode
53	Check value

Note 1: xx/xx data corresponds to 57ES/42ES parameters.

Note 2: If the baud rate, address, and packet address parameters are 0, then the original parameters are retained.



4.7.3 Command to read all status parameters

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	48H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte 4-Byte 23	Byte24
Header	Address	Code	Parameter value	Checksum
FB	addr	48H	params	CRC

The values for the parmas parameter are defined in the table below.

Read all status parameters (params)		
byte	Return data format	Corresponding single instruction
1	FB	
2	1	
3	48H	
4	Motor status	F1
5	Enable state	3A
6	stalled state	3E
7	IO status	34
8	Encoder value	31
9		
10		
11		
12		
13		
14	Real-time speed	32
15		
16	pulse count	33
17		
18		
19		
20	Angular error	39
21		
22		
23		
24	Check value	



Part 5. IO Port Operation Instructions

5.1 Read I/O port status

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	34H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Port status	Checksum
FB	addr	34H	status (uint8_t)	CRC

Read port status definition table

status							
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Undefined				ALM	PEND	IN_2	IN_1

PEND 1: In place; 0: Not in place

ALM 1: No alarm; 0: Alarmed

Note: 42ES does not have the IN_2 port; bit1 corresponds to the state of the En port.



5.2 Write data to IO port

The command to write a port is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Port data	Checksum
FA	addr	36H	data	CRC

Write port status definition table

data							
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
alm_mask		pend_mask		ALM	PEND	0	0

alm_mask 0: Do not write to the ALM IO port
1: Write the ALM value to the ALM IO port
2: The ALM IO port value remains unchanged.
pend_mask 0: Do not write to the PEND IO port
1: Write the PEND value to the PEND IO port.
2: The PEND IO port value remains unchanged.
ALM ALM port write value (0/1)
PEND PEND port writes value (0/1)

Note: Writing 1 to the ALM IO port disconnects the corresponding optocoupler.

Writing 1 to the PEND IO port closes the corresponding optocoupler.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	36H	status (uint8_t)	CRC

status = 0 Failed to write to port
status = 1 Successfully written to port



5.3 IO port pulse frequency division output

Map the PEND port to a pulse divider output port.

The configuration command is as follows:

Byte1	Byte2	Byte3	Byte4	Bytes5-8	Bytes9
Header	address	Code	Start level	Frequency division period	Checksum
FA	addr	9F	divLevel	divPeriod	CRC

divLevel 0: Low start level (default)

 1: High start level

divPeriod Frequency division period (default 0)

When divPeriod < 100, no frequency divider output

When divPeriod ≥ 100, the PEND port toggles once every divPeriod pulse cycle.

For example, setting 16 subdivisions, divPeriod = 3200, then the PEND port flips once for every revolution of the motor.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	9FH	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.

Note: To disable this feature, simply set divPeriod = 0.

Part 6. Motor return to zero instructions

There are two types of motor homing methods: "origin homing" and "coordinate homing".

6.1 Explanation of the method of returning to zero at the origin

There are two ways to "return to zero": "return to zero by origin switch" and "return to zero by mechanical limit switch".

1. Return the origin switch to zero.

Connect the origin switch to the corresponding port:

Origin switch access port	
SERVO57ES	IN_COM / IN1
SERVO42ES	IN+ / IN-

For detailed wiring instructions, please refer to 2.4External switch wiring method".

The process of returning to zero is as follows:

- The motor first searches for the switch in the set "direction" and "high speed " ;
- Upon encountering the rising edge of the switch signal, it begins to operate at "low speed " and disengages from the switch;
- After the switch is disconnected, the motor reverses direction and continues to search for the rising edge of the switch signal at "low speed " ;
- Upon encountering the rising edge of the switch signal, it then runs at "high speed " to the preset "origin offset" position and stops.
- Mark the current position as "coordinate zero point", and the origin has successfully returned to zero.

2. Mechanical limit switch returns to zero.

The zero-return torque when the mechanical limit is returned to zero needs to be set in advance via command 94H. The set torque should be sufficient to drive the load, but should not be too large to avoid damaging the equipment.

The process of returning to zero is as follows:

- searches for the mechanical limit position using the set "direction", "torque " , and "low speed " ;
- When encountering a mechanical limit switch, the rotor will stop and then run to the preset "origin offset" position to stop.
- Mark the current position as "coordinate zero point", and the origin has successfully returned to zero.



3. Single lap to zero

The "zero point" of the coordinates within a single circle needs to be set in advance via command 92H.

The direction of the single-turn zeroing needs to be set in advance via command 93H: "forward", "reverse", or "nearest".

The process of returning to zero is as follows:

- The motor returns to the "coordinate zero point" position within a preset single revolution at high speed in the set zero-return direction;
- Upon arrival, the current position is reset to zero, and the single-lap coordinates have successfully returned to zero.

6.2 Explanation of coordinate zeroing method

To return to zero directly, you need to first execute the "Return to Zero" function to determine the "zero point".

The process of returning to zero is as follows:

No search process is required; it directly and quickly reaches the "coordinate zero point" position and successfully returns to zero.

6.3 Set parameters related to zero return.

6.3.1 Set the zero-return mode, direction, and speed parameters.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)							
Byte1	Byte2	Byte3	Byte4	Byte5	Bytes6-7	Bytes8-9	Bytes10
Header	Address	Code	mode	direction	"High speed"	"Low speed"	Checksum
FA	addr	95H	HmMode	HmDir	HiSpeed	LoSpeed	CRC

Hm Mode: Sets the zero-return mode

00: Origin switch returns to zero (default value)

01: Mechanical limit switch returns to zero

02: Single lap zeroing

03: Disable the zero-reset function

Hm Dir: Sets the return-to-zero direction

00: Positive zero return (default value)

01: Reverse return to zero

02: Zero at the nearest point (only applicable to zeroing on a single lap)

HiSpeed: Sets the "high-speed" RPM for homing (range 1~3000 RPM, default = 100).

LoSpeed: Sets the "low speed" RPM for homing(range 1~100RPM, default = 10).

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	95H	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.



6.3.2 Set the zero-return torque and origin offset parameters.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)					
Byte1	Byte2	Byte3	Bytes4-5	Byte6-9	Bytes10
Header	Address	Code	Zero-return current	Origin offset	Checksum
FA	addr	96H	HmA Torque (uint16_t)	Orgoffset (int32_t)	CRC

HmA Torque : Sets the current when the mechanical limit switch returns to zero.

SERV042ES maximum return-to-zero current is 3000mA (default 300mA).

SERV057ES maximum return-to-zero current is 5200mA (default 600mA).

Orgoffset: Sets the offset position after returning to zero (default 0x2000).

When Orgoffset = 0, the motor returns to zero and then stops.

When Orgoffset ≠ 0, after the motor returns to zero, it will continue running until it reaches the offset position before stopping.

Note: The Orgoffset value is invalid when lapping back to zero.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	96H	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.

6.3.3 Configure the zero-return trigger method and timeout parameters.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)					
Byte1	Byte2	Byte3	Byte4	Bytes 5-8	Byte9
Header	Address	Code	trigger method	Zero-return timeout(ms)	Checksum
FA	addr	97H	HmTrig	HmTimOut(uint32_t)	CRC

HmTrig: Sets the zero-return trigger method

00: Command triggers reset to zero (default value)

01: Automatic return to zero upon power-on (only valid in bus mode)

02: En signal triggers zero return (only valid in pulse control mode)

In pulse control mode, when hmTrig = 2, the motor is triggered to return to zero when the En signal line generates a 200ms-wide disabled level. (Pulse recognition range: 150ms~250ms)

When En is low, it enables the motor and generates a 200ms high-level signal, triggering the motor to return to zero.

When En is high, it enables the motor and generates a 200ms low-level signal, triggering the motor to return to zero.

HmTimOut: Sets the timeout period for returning to zero (default = 120000).

If the motor does not reach zero within this time, a timeout failure message will be returned.



Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	97H	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.

6.3.4 Setting Zero Point Instructions

This command directly sets the current position to "zero".

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	CMD	Checksum
FA	addr	92H	00H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	92H	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.

6.3.5 Execute the zero-return instruction

The execution instructions are as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	method	Checksum
FA	addr	91H	goZeroMode	CRC

goZeroMode 00: Executes the "Return to Zero" function.

 01: Execute the "Coordinate Zeroing" function

Note: You can execute "Return to Zero" first to determine the "Zero Point Coordinates" before executing "Return to Zero".

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	91H	status (uint8_t)	CRC

status = 0 failed to return to zero

status = 1 starting return to zero

status = 2 return to zero complete

status = 3 times out and failed

6.4 Example of Switch return to zero configuration

Connect the origin switch to the corresponding port, referring to "

6.1 Explanation of the method of returning to zero at the origin"

6.4.1 Example of a switch command triggering a return to zero

The setup steps are as follows:

1. Set **working mode**
FA 01 82 05 82
2. Set the switch **trigger level**
FA 01 9E FF 00 FF 97
3. Set the return-to-zero **mode**, **direction** , "High Speed" , "Low Speed".
FA 01 9 5 00 00 00 64 00 0A FE
4. Set **the origin offset**
FA 01 9 6 02 58 00 00 20 00 0B
5. Configure the zero-return **trigger method** and **timeout parameters**.
FA 01 9 7 00 00 00 EA 60 DC
6. Save settings
FA 01 60 01 5C
7. Perform "return to zero"
FA 01 91 00 8C

It can be observed that: the motor rotates forward at high speed (100RPM) → the switch is triggered → the motor rotates in reverse at low speed (10RPM) → the motor rotates forward at low speed (10RPM) → the switch is triggered again → the motor runs at high speed to the offset position and returns to zero.

```
[2025-10-15 14:07:19.638]# SEND HEX>
FA 01 82 05 82
[2025-10-15 14:07:19.704]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 14:07:26.302]# SEND HEX>
FA 01 9E FF 00 FF 97
[2025-10-15 14:07:26.372]# RECV HEX>
FB 01 9E 01 9B
[2025-10-15 14:07:32.638]# SEND HEX>
FA 01 95 00 00 00 64 00 0A FE
[2025-10-15 14:07:32.696]# RECV HEX>
FB 01 95 01 92
[2025-10-15 14:07:38.448]# SEND HEX>
FA 01 96 02 58 00 00 20 00 0B
[2025-10-15 14:07:38.524]# RECV HEX>
FB 01 96 01 93
[2025-10-15 14:07:45.608]# SEND HEX>
FA 01 97 00 00 00 EA 60 DC
[2025-10-15 14:07:45.682]# RECV HEX>
FB 01 97 01 94
[2025-10-15 14:07:51.774]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 14:07:51.834]# RECV HEX>
FB 01 60 01 5D
[2025-10-15 14:07:57.678]# SEND HEX>
FA 01 91 00 8C
[2025-10-15 14:07:57.741]# RECV HEX>
FB 01 91 01 8E
[2025-10-15 14:08:15.869]# RECV HEX>
FB 01 91 02 8F
```



6.4.2 Example of automatic return to zero upon power-on

The setup steps are as follows:

1. Set **working mode**
FA 01 82 05 82
2. Set the switch **trigger level**
FA 01 9E FF 00 FF 97
3. Set the **return-to-zero mode, direction, "High Speed", "Low Speed"**.
FA 01 95 00 00 00 64 00 0A FE
4. Set **the origin offset**
FA 01 96 02 58 00 00 20 00 0B
5. Configure the zero-return **trigger method** and **timeout parameters**.
FA 01 97 01 00 00 EA 60 DD
6. **Save settings**
FA 01 60 01 5C

After a power outage and restart, the following can be observed: the motor rotates at high speed (100 RPM) in the forward direction → the switch is triggered → the motor rotates in the reverse direction at low speed (10 RPM) → the motor rotates in the forward direction at low speed (10 RPM) → the switch is triggered again → the motor runs at high speed to the offset position and returns to zero.

```
[2025-10-15 14:10:24.038]# SEND HEX>
FA 01 82 05 82
[2025-10-15 14:10:24.090]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 14:10:29.206]# SEND HEX>
FA 01 9E FF 00 FF 97
[2025-10-15 14:10:29.271]# RECV HEX>
FB 01 9E 01 9B
[2025-10-15 14:10:34.678]# SEND HEX>
FA 01 95 00 00 00 64 00 0A FE
[2025-10-15 14:10:34.746]# RECV HEX>
FB 01 95 01 92
[2025-10-15 14:10:41.166]# SEND HEX>
FA 01 96 02 58 00 00 20 00 0B
[2025-10-15 14:10:41.238]# RECV HEX>
FB 01 96 01 93
[2025-10-15 14:10:47.414]# SEND HEX>
FA 01 97 01 00 00 EA 60 DD
[2025-10-15 14:10:47.489]# RECV HEX>
FB 01 97 01 94
[2025-10-15 14:10:52.990]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 14:10:53.079]# RECV HEX>
FB 01 60 01 5D
```

Note: If the limit switch is not triggered within the set "return to zero timeout period", the "return to zero timeout failure" message will be returned.

6.5 Example of mechanical limit switch homing configuration

A suitable zero-return current needs to be set for mechanical limit switches. The zero-return current should not be too large, just enough to drive the load, to avoid damaging the mechanical device.

6.5.1 Example of mechanical limit command triggering return to zero

1. Set **working mode**

FA 01 82 05 82

2. Set the **return-to-zero mode, direction, "High Speed", "Low Speed"**.

FA 01 95 01 00 00 64 00 0A FF

3. Set **the zero-return current** and **the origin offset**.

FA 01 96 02 58 00 00 20 00 0B

4. Configure the **zero-return trigger method** and **timeout parameters**.

FA 01 9 7 00 00 00 EA 60 DC

5. **Save settings**

FA 01 60 01 5C

6. Perform **"return to zero"**

FA 01 91 00 8C

It can be observed that: the motor rotates forward at low speed (10RPM) → after touching the mechanical limit switch → the motor stops → the motor runs in reverse at high speed to the offset position, and the return to zero is completed.

```
[2025-10-15 14:19:13.235]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 14:19:18.382]# SEND HEX>
FA 01 95 01 00 00 64 00 0A FF
[2025-10-15 14:19:18.453]# RECV HEX>
FB 01 95 01 92
[2025-10-15 14:19:23.799]# SEND HEX>
FA 01 96 02 58 00 00 20 00 0B
[2025-10-15 14:19:23.868]# RECV HEX>
FB 01 96 01 93
[2025-10-15 14:19:29.406]# SEND HEX>
FA 01 97 00 00 00 EA 60 DC
[2025-10-15 14:19:29.467]# RECV HEX>
FB 01 97 01 94
[2025-10-15 14:19:37.487]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 14:19:37.559]# RECV HEX>
FB 01 60 01 5D
[2025-10-15 14:19:44.101]# SEND HEX>
FA 01 91 00 8C
[2025-10-15 14:19:44.171]# RECV HEX>
FB 01 91 01 8E
[2025-10-15 14:19:51.033]# RECV HEX>
FB 01 91 02 8F |
```

Note: If the motor does not touch the mechanical limit within the set "return to zero timeout period", a "return to zero timeout failure" message will be returned.



6.5.2 Example of automatic zeroing upon power-on of mechanical limit switch

1. Set **working mode**
FA 01 82 05 82
2. Set the **return-to-zero mode** , **direction** : " **High Speed** " or " **Low Speed** ".
FA 01 95 01 00 00 64 00 0A FF
3. Set **the zero-return current** and **the origin offset**.
FA 01 96 02 58 00 00 20 00 0B
4. Configure the zero-return **trigger method** and **timeout parameters**.
FA 01 97 01 00 00 EA 60 DD
5. **Save settings**
FA 01 60 01 5C

After a power outage and restart, the following can be observed: the motor rotates forward at low speed (10 RPM) → after touching the mechanical limit switch → the motor stops → the motor runs in reverse at high speed to the offset position, and the return to zero is completed.

```
[2025-10-15 14:24:25.287]# SEND HEX>
FA 01 82 05 82
[2025-10-15 14:24:25.347]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 14:24:30.358]# SEND HEX>
FA 01 95 01 00 00 64 00 0A FF
[2025-10-15 14:24:30.427]# RECV HEX>
FB 01 95 01 92
[2025-10-15 14:24:35.918]# SEND HEX>
FA 01 96 02 58 00 00 20 00 0B
[2025-10-15 14:24:35.973]# RECV HEX>
FB 01 96 01 93
[2025-10-15 14:24:41.910]# SEND HEX>
FA 01 97 01 00 00 EA 60 DD
[2025-10-15 14:24:41.970]# RECV HEX>
FB 01 97 01 94
[2025-10-15 14:24:46.902]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 14:24:46.970]# RECV HEX>
FB 01 60 01 5D
```

Note: If the motor does not touch the mechanical limit within the set "return to zero timeout period", a "return to zero timeout failure" message will be returned.



6.6 Example of Single-lap zero-return configuration

To return to zero in a single lap, the "coordinate zero point" within the single lap must be set in advance using command 92H.

6.6.1 Example of single-lap command triggering zero return

First, move the motor shaft to the appropriate "zero point" position.

1. Set **working mode**

FA 01 82 05 82

2. Set the **return-to-zero mode**, "**Nearest**", "**High Speed**", "**Low Speed**"

FA 01 95 02 02 00 64 00 0A 02 (Return to zero at the nearest position)

3. Configure the **zero-return trigger method** and **timeout parameters**.

FA 01 97 00 00 00 EA 60 DC

4. Set the **"zero point"** coordinates

FA 01 92 00 8D

5. **Save settings**

FA 01 60 01 5C

6. Perform **"return to zero"**

FA 01 91 00 8C

After saving the settings, you can move the motor shaft away from the "zero" position by turning off the power or loosening the shaft.

After executing "return to zero", it can be observed that the motor runs at high speed (100RPM) to the "zero point" position, and the return to zero is completed.

```
[2025-10-15 14:49:49.511]# SEND HEX>
FA 01 82 05 82
[2025-10-15 14:49:49.571]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 14:49:54.607]# SEND HEX>
FA 01 95 02 02 00 64 00 0A 02
[2025-10-15 14:49:54.663]# RECV HEX>
FB 01 95 01 92
[2025-10-15 14:50:00.847]# SEND HEX>
FA 01 97 00 00 00 EA 60 DC
[2025-10-15 14:50:00.916]# RECV HEX>
FB 01 97 01 94
[2025-10-15 14:50:06.640]# SEND HEX>
FA 01 92 00 8D
[2025-10-15 14:50:06.711]# RECV HEX>
FB 01 92 01 8F
[2025-10-15 14:50:21.375]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 14:50:21.445]# RECV HEX>
FB 01 60 01 5D
[2025-10-15 14:51:19.671]# SEND HEX>
FA 01 91 00 8C
[2025-10-15 14:51:19.725]# RECV HEX>
FB 01 91 01 8E
[2025-10-15 14:51:21.012]# RECV HEX>
FB 01 91 02 8F
```



6.6.2 Example of automatic zero return upon power-on in a single revolution

First, move the motor shaft to the appropriate "zero point" position.

1. Set **working mode**

FA 01 82 05 82

2. Set the return-to-zero mode, "Nearest", "High Speed", "Low Speed"

FA 01 95 02 02 00 64 00 0A 02 (Return to zero at the nearest position)

3. Configure the zero-return trigger method and timeout parameters.

FA 01 97 01 00 00 EA 60 DD

4. Set the "zero point" coordinates

FA 01 92 00 8D

5. Save settings

FA 01 60 01 5C

After power is cut off, move the motor shaft away from the "zero point" position.

After powering on again, it can be observed that the motor runs at high speed (100RPM) to the "zero point" position and returns to zero.

```
[2025-10-15 14:56:43.871]# SEND HEX>
FA 01 82 05 82
[2025-10-15 14:56:43.939]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 14:56:50.887]# SEND HEX>
FA 01 95 02 02 00 64 00 0A 02
[2025-10-15 14:56:50.964]# RECV HEX>
FB 01 95 01 92
[2025-10-15 14:56:56.911]# SEND HEX>
FA 01 97 01 00 00 EA 60 DD
[2025-10-15 14:56:56.987]# RECV HEX>
FB 01 97 01 94
[2025-10-15 14:57:05.631]# SEND HEX>
FA 01 92 00 8D
[2025-10-15 14:57:05.703]# RECV HEX>
FB 01 92 01 8F
[2025-10-15 14:57:12.704]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 14:57:12.754]# RECV HEX>
FB 01 60 01 5D
```

6.7 Example of coordinate zeroing

Before performing "coordinate zeroing", the coordinates of the "zero point" must be determined:

For switch-based or mechanical limit switch-based zeroing, "origin zeroing" must be performed first.

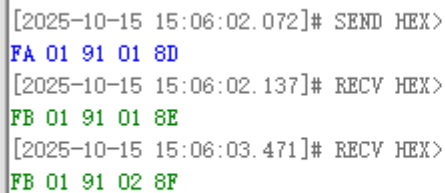
For the single-lap zero-return method, simply set the "zero point" coordinates.

1. Execute "Coordinate Zeroing"

FA 01 91 01 8D

Once the "zero point" coordinates are determined, regardless of the motor's position, executing "coordinate return to zero" will cause the motor to return to the zero point position at high speed.

Note: The motor must be enabled before executing "coordinate return to zero".



```
[2025-10-15 15:06:02.072]# SEND HEX>  
FA 01 91 01 8D  
[2025-10-15 15:06:02.137]# RECV HEX>  
FB 01 91 01 8E  
[2025-10-15 15:06:03.471]# RECV HEX>  
FB 01 91 02 8F
```



Part 7. Left and right limit switch instructions

The default limit switch connection method is shown in the table below:

	Left limit port	Right limit port	Remark
SERV057ES	IN_COM / IN1	IN_COM / IN2	
SERV042ES	IN+ / IN-	EN+/EN-	Bus mode, enable right limit switch

7.1 Set limit switch parameters

The limit switch setting instructions are as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
Header	Address	Code	Limit enable	Trigger level	Limit mapping	Checksum
FA	addr	9EH	EndLimit	EndLevel	reMap	CRC

EndLimit limit enable

00 : Limit switch function disabled (default value)

01 : Enable limit function

FF: Keep the original settings unchanged.

EndLevel Effective level when the End Level switch is closed

00 : Low level when the switch is closed (default value)

01 : High level when the switch is closed

FF: Keep the original settings unchanged.

reMap 00: Disable right limit port mapping (default value, only valid for 42ES)

01: Enable right limit port mapping

FF: Keep the original settings unchanged.

Limit enable description:

1. When EndLimit = 1, in bus control mode

Left limit switch triggered, motor stops moving to the left.

Right limit switch triggered, motor stops moving to the right.

2. After using the limit function for the first time or changing the limit parameters, **you must perform a "return to zero" operation once**.

3. The limit function is invalid in pulse control mode.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	9EH	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.



7.2 Example of Limit switch configuration

Connect the limit switch to the corresponding port.

1. Set **working mode**

FA 01 82 05 82

2. Set **limit enable**, **trigger level**, and **right limit mapping**.

57ES Command: FA 01 9E 01 00 FF 99 (No right limit mapping required)

42ES Command: FA 01 9E 01 00 01 9B (Right limit mapping required)

3. Set the **return-to-zero mode**, **direction**, "**High Speed**", "**Low Speed**".

FA 01 95 00 00 00 64 00 0A FE

4. Perform "return to zero"

FA 01 91 00 8C

(After using the limit function for the first time or changing the limit switch parameters, you must perform a "zeroing" operation once .)

5. Save settings

FA 01 60 01 5C

```
[2025-10-16 08:57:28.366]# SEND HEX>
FA 01 82 05 82
[2025-10-16 08:57:28.422]# RECV HEX>
FB 01 82 01 7F
[2025-10-16 08:57:39.102]# SEND HEX>
FA 01 9E 01 00 FF 99
[2025-10-16 08:57:39.174]# RECV HEX>
FB 01 9E 01 9B
[2025-10-16 08:57:44.726]# SEND HEX>
FA 01 95 00 00 00 64 00 0A FE
[2025-10-16 08:57:44.794]# RECV HEX>
FB 01 95 01 92
[2025-10-16 08:57:51.814]# SEND HEX>
FA 01 91 00 8C
[2025-10-16 08:57:51.880]# RECV HEX>
FB 01 91 01 8E
[2025-10-16 08:58:05.384]# RECV HEX>
FB 01 91 02 8F
[2025-10-16 08:58:14.862]# SEND HEX>
FA 01 60 01 5C
[2025-10-16 08:58:14.914]# RECV HEX>
FB 01 60 01 5D
```



7.3 Example of Left Limit switch test

1. Left limit switch test

Controlling motor operation

FA 01 FD 81 2C 02 0F 00 00 00 B6

When the motor is running, triggering the left limit switch will stop the motor.

```
[2025-10-16 08:58:22.605]# SEND HEX>
FA 01 FD 81 2C 02 0F 00 00 B6
[2025-10-16 08:58:22.674]# RECV HEX>
FB 01 FD 01 FA
[2025-10-16 08:58:29.900]# RECV HEX>
FB 01 FD 03 FC
```

7.4 Example of Right Limit switch test

1. Right limit switch test

Controlling motor operation

FA 01 FD 01 2C 02 0F 00 00 00 36

When the motor is running, triggering the right limit switch will stop the motor.

```
[2025-10-16 08:59:15.110]# SEND HEX>
FA 01 FD 01 2C 02 0F 00 00 00 36
[2025-10-16 08:59:15.189]# RECV HEX>
FB 01 FD 01 FA
[2025-10-16 08:59:21.817]# RECV HEX>
FB 01 FD 03 FC
```



Part 8. Bus control mode parameter description

Note 1 : The instructions in this section are only valid in bus control mode.

Note 2: MODBUS-RTU protocol instructions, see Part 7.

8.1 Description the parameters of speed and acceleration

1.speed

The speed parameter ranges from 0 to 3000. The larger the value, the faster the motor rotates.

When speed = 0, the motor stops rotating.

If the set speed is greater than the maximum speed of the control mode, the motor runs at the maximum speed of the control mode.

Note: The speed value is calibrated based on 16/32/64 subdivisions, and the speeds of other subdivisions need to be calculated based on 16 subdivisions.

For example, setting speed=1200

At 8 subdivisions, the speed is 2400 (RPM)

At 16/32/64 subdivisions, the speed is 1200 (RPM)

At 128 subdivisions, the speed is 150 (RPM)



2. acceleration

The value of the acceleration(acc) ranges from 0 to 255. The larger the value, the faster the motor accelerates/decelerates.

If acc=0, the motor runs without acceleration or deceleration, and runs directly at the set speed.

① accelerates

Suppose at time t_1 , the current speed is V_{t1} ($V_{t1} < \text{speed}$)

at time t_2 , the current speed is V_{t2}

$$t_2 - t_1 = (256 - \text{acc}) * 50 (\mu\text{S})$$

The relationship between the current speed V_{ti} , acc, and speed is as follows:

$$V_{t2} = V_{t1} + 1 \quad (V_{t2} \leq \text{speed})$$

For example: acc = 236, speed = 3000

T(ms)	speed (RPM)
0	0
1	1
2	2
3	3
...	...

T(ms)	speed (RPM)
...	...
...	...
2998	2998
2999	2999
3000	3000

② decelerates

Suppose at time t_1 , the current speed is V_{t1} ($V_{t1} > \text{speed}$)

at time t_2 , the current speed is V_{t2}

$$t_2 - t_1 = (256 - \text{acc}) * 50 (\mu\text{S})$$

The relationship between the current speed V_{ti} , acc, and speed is as follows:

$$V_{t2} = V_{t1} - 1 \quad (V_{t2} \geq \text{speed})$$



8.2 Bus control general instructions

8.2.1 Read motor operating status

The read command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	F1H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Motor operating status	Checksum
FB	addr	F1H	status	CRC

status: Motor operating status, corresponding codes are shown in the table below.

Motor operating status	status	Remark
Query failed	0	
Stop running	1	
Speed up	2	
slow down	3	
Run at full speed	4	
Reset to zero	5	

Note : This instruction is only valid in bus control mode.



8.2.2 Set motor enable state

In bus control mode, the motor's enable state is no longer controlled by the level of the En pin, but is controlled by the command.

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Enable	Checksum
FA	addr	F3H	enable	CRC

enable = 00 Motor is disabled (loose shaft).

enable = 01 Motor enable (shaft lock)

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	F3H	status (uint8_t)	CRC

status = 0 set failed.

status = 1 set succeeded.

Note: This instruction is only valid in bus control mode.

8.2.3 Motor emergency stop command

The emergency stop command is as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	addr	F7H	CRC

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	state	Checksum
FB	addr	F7H	status (uint8_t)	CRC

status = 0 Emergency stop failed

status = 1 Emergency stop successful

Note: Emergency stop commands are not recommended when the motor speed exceeds 1000 RPM !



Part 9. Speed Control Mode Description

In speed control mode,

It can control the motor to run continuously at a set acceleration and speed.

It can control the motor to run at a set acceleration and speed for a set period of time.

It allows the motor to run at a set speed and acceleration as soon as it is turned on.

9.1 Speed control mode operation instructions

The execution command format is as follows:

Downlink frames (PC → SERV0xxES)							
Byte1	Byte2	Byte3	Byte4		Byte5	Byte6	Byte7
Header	Address	Code	direction	REV	speed		ACC
FA	addr	F6H	b7 dir	b6-b4 - -	b3 -b0 speed	b7-b0 acc	CRC

Byte4: The highest bit indicates direction, and the lower 4 bits, together with byte5, indicate speed.

Byte5: Byte5 and the lower 4 bits of Byte4 together represent the speed.

The parameters are explained below:

dir value range 0 - 1 (CCW/CW)

speed value range 0-3000

acc values range from 0 to 255.

for example:

Send FA 01 F6 02 80 02 75 , the motor rotates forward at an acc=2 and a speed = 640 (RPM).

Send FA 01 F6 82 80 02 F5 , the motor reverses at an acc=2 and speed=640 (RPM).

If you want the motor to automatically stop after running for a period of time, the operating command is as follows:

Downlink frames (PC → SERV0xxES)								
Byte1	Byte2	Byte3	Byte4		Byte5	Byte6	Bytes7-10	bytes11
Header	address	Code	direction	reserve	speed		ACC	runtime
FA	addr	F6H	b7 dir	b6-b4 - -	b3-b0 speed	b7-b0 acc	acc	runTime
								CRC

Run time unit: 10ms , range (0 ~ 19990000H)



Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	F6H	status (uint8_t)	CRC

status = 0 Running failed

status = 1 Start running

status = 2 Running completed

status = 5 Synchronous mode has received the instruction
(it will only start running after receiving the synchronization
flag).

Note: This instruction is only valid in bus control mode.

9.2 stop command

The stop command is as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
Header	Address	Code	control word	control word	acceleration	Checksum
FA	addr	F6H	00	00	acc	CRC

The stop command can control the motor to slow down and stop gradually, or it can control the motor to stop immediately. When acc is set to $\neq 0$, the motor decelerates and stops slowly. When acc is set to $= 0$, the motor stops immediately.

① Slow down and stop command (acc $\neq 0$)

for example:

Send FA 01 F6 0 0 00 02 F3

stop rotating at a deceleration of acc=2.

② Immediate stop command (acc = 0)

for example:

Send FA 01 F6 0 0 00 00 F1

Stop the motor immediately

Note: It is not recommended to use the immediate stop command when the motor speed exceeds 1000 RPM !

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	F6H	status (uint8_t)	CRC

status = 0 Stop failure

status = 1 Stop starting

status = 2 Stopped completing

status = 3 Stopped upon touching limit switch

Note: You can also use the emergency stop command F7H to stop the operation.



9.3 Set automatic start command upon power-on

After setting automatic operation, the motor will automatically run in the set direction, speed, and acceleration every time the machine is turned on.

The format for setting the automatic operation command upon power-on is as follows:

Downlink frames (PC → SERV0xxES)								
Byte1	Byte2	Byte3	Byte4			Byte5	Byte6	Byte7
Header	Address	Code	direction	REV	speed		ACC	Checksum
FA	addr	FFH	b7	b6-b4	b 3 -b0	b7-b0	acc	CRC
			dir	- -	speed			

dir Direction value range 0-1 (default 0)
speed speed value range 0-3000 (default 0)
acc acceleration value range 0-255 (default 0).

Note: When speed=0, automatic operation upon power-on is disabled.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	FFH	status (uint8_t)	CRC

status = 0 set failed.
status = 1 set succeeded.



9.4 Example of Speed mode running

1. Set working mode

FA 01 82 05 82

2. Save settings

FA 01 60 01 5C

3. Set the running direction, speed, and acceleration parameters.

FA 01 F6 01 2C 02 20

At this point, the motor can be observed to run in the set direction, speed, and acceleration.

To stop the operation, execute the stop command.

4. Stop command

FA 01 F6 00 00 02 F3

At this point, the motor can be observed to stop at the set acceleration.

```
[2025-11-21 11:04:30.525]# SEND HEX>
FA 01 82 05 82
[2025-11-21 11:04:30.585]# RECV HEX>
FB 01 82 01 7F
[2025-11-21 11:04:35.560]# SEND HEX>
FA 01 60 01 5C
[2025-11-21 11:04:35.605]# RECV HEX>
FB 01 60 01 5D
[2025-11-21 11:04:40.736]# SEND HEX>
FA 01 F6 01 2C 02 20
[2025-11-21 11:04:40.786]# RECV HEX>
FB 01 F6 01 F3
[2025-11-21 11:04:49.435]# SEND HEX>
FA 01 F6 00 00 02 F3
[2025-11-21 11:04:49.500]# RECV HEX>
FB 01 F6 01 F3
[2025-11-21 11:04:52.991]# RECV HEX>
FB 01 F6 02 F4
```



9.5 Example of Automatic run mode

1. Set working mode

FA 01 82 05 82

2. Set params of automatically run upon power-on

FA 01 FF 01 2C 02 29

3. Save settings

FA 01 60 01 5C

the motor will run automatically after a power outage and restart.

```
[2025-11-21 11:06:46.134]# SEND HEX>
FA 01 82 05 82
[2025-11-21 11:06:46.199]# RECV HEX>
FB 01 82 01 7F
[2025-11-21 11:06:52.048]# SEND HEX>
FA 01 FF 01 2C 02 29
[2025-11-21 11:06:52.118]# RECV HEX>
FB 01 FF 01 FC
[2025-11-21 11:06:57.029]# SEND HEX>
FA 01 60 01 5C
[2025-11-21 11:06:57.099]# RECV HEX>
FB 01 60 01 5D |
```

To cancel automatic operation upon power-on, execute the following command.

1. Stop command

FA 01 F6 00 00 02 F3

2. Clear params of automatically run upon power-on

FA 01 FF 00 00 00 FA

3. Save settings

FA 01 60 01 5C

After a power outage and restart, the motor will no longer run automatically.

```
[2025-11-21 11:08:39.532]# SEND HEX>
FA 01 F6 00 00 02 F3
[2025-11-21 11:08:39.597]# RECV HEX>
FB 01 F6 01 F3
[2025-11-21 11:08:43.078]# RECV HEX>
FB 01 F6 02 F4
[2025-11-21 11:08:46.023]# SEND HEX>
FA 01 FF 00 00 00 FA
[2025-11-21 11:08:46.093]# RECV HEX>
FB 01 FF 01 FC
[2025-11-21 11:08:53.014]# SEND HEX>
FA 01 60 01 5C
[2025-11-21 11:08:53.089]# RECV HEX>
FB 01 60 01 5D |
```



Part 10. Position control mode description

10.1 Relative motion according to pulse number

In this mode, the motor can be controlled to run to a specified position relative to the set acceleration and speed, based on the number of pulses.

10.1.1 Pulse relative operation command

The execution command format is as follows:

Downlink frames (PC → SERV0xxES)								
Byte1	Byte2	Byte3	Byte4		Byte5	Byte6	Bytes7-10	Bytes11
Header	Addr	Code	direction	reserve	speed		ACC	pulses
FA	addr	FD	b7 dir	b6-b4 - -	b3 -b0 speed	b7-b0	acc	relPulses CRC

Byte4: The highest bit indicates direction, and the lower 4 bits, together with byte5, indicate speed.

Byte5: Byte5 and the lower 4 bits of Byte4 together represent the speed.

The parameters are explained below:

dir Direction, value range 0 - 1 (CCW/CW)

Speed Speed, value range 0 - 3000 (RPM)

Acc Accelerator value range 0 - 255.

relPulses Pulses count value range 0 - 0xFFFFFFFF

for example:

Send FA 01 FD 02 80 02 00 00 FA 00 76 , the motor will rotate 20 revolutions in the forward direction (16 microsteps) with acc =2 and speed=0x280 ;

Send FA 01 FD 82 80 02 00 00 FA 00 F6 , the motor rotates in the opposite direction for 20 revolutions (16 microsteps) with acc=2 and speed=0x280 ;

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	FDH	status (uint8_t)	CRC

status = 0 start failed

status = 1 start running

status = 2 position completed

status = 3 stopped upon touching limit switch

status = 5 synchronous mode has received the instruction (it will only start running after receiving the synchronization flag).



10.1.2 Stop command

The stop command is as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Bytes 4-5	Byte6	Bytes7-10	bytes11
Header	Address	Code	data	acceleration	data	Checksum
FA	addr	FDH	00	acc	00	CRC

The stop command can control the motor to slow down and stop gradually, or it can control the motor to stop immediately. When acc is set to $\neq 0$, the motor decelerates and stops slowly. When acc is set to $= 0$, the motor stops immediately.

① Slow down and stop command (acc $\neq 0$)

for example:

Send FA 01 FD 00 00 02 00 00 00 00 FA
stop rotating at a deceleration of acc=2.

② Immediate stop command (acc = 0)

for example:

Send FA 01 FD 00 00 00 00 00 00 00 F8
Stop the motor immediately

Note: It is not recommended to use the immediate stop command when the motor speed exceeds 1000 RPM !

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	FDH	status (uint8_t)	CRC

Status = 0	Stop failed
status = 1	Stop Start
status = 2	position stopped completing
status = 3	Stopped upon touching limit switch

Note: You can also use the emergency stop command F7H to stop the operation.



10.1.3 Example of Pulse Relative Operation

1. Set working mode
FA 01 82 05 82
2. Save settings
FA 01 60 01 5C
3. Set the running direction, speed, and relative pulse count.
FA 01 FD 01 2C 02 00 04 E2 00 0D
(At speed = 300 RPM , acceleration = 2, 100 revolutions clockwise
(16 subdivisions))
4. If you need to slow down and stop the machine during operation,
the command is as follows:
FA 01 FD 00 00 02 00 00 00 00 FA
5. If an emergency stop is required during operation, the command is
as follows:
FA 01 F7 F2

```
[2025-10-15 16:25:55.345]# SEND HEX>
FA 01 82 05 82
[2025-10-15 16:25:55.417]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 16:26:00.114]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 16:26:00.189]# RECV HEX>
FB 01 60 01 5D
[2025-10-15 16:26:11.178]# SEND HEX>
FA 01 FD 01 2C 02 00 04 E2 00 0D
[2025-10-15 16:26:11.244]# RECV HEX>
FB 01 FD 01 FA
[2025-10-15 16:26:34.555]# RECV HEX>
FB 01 FD 02 FB
```



10.2 Absolute motion based on pulse count

In this mode, the motor can be controlled to run to the specified position with a set acceleration and speed, based on the number of pulses.

Note: Before using absolute motion, a zeroing process must be performed once (instruction 0x91 or 0x92) to mark the "zero point".

10.2.1 Pulse Absolute Operation Command

The execution command format is as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Byte4-5	Byte6	Bytes7-10	Bytes11
Header	Address	Code	speed	acceleration	pulses count	Checksum
FA	addr	FEH	speed	acc	absPulses	CRC

The parameters are explained below:

Speed speed value range 0 - 3000 (RPM)

Acc accelerator value range 0 - 255.

absPulses absolute number of pulses,
value range of int32_t (-2147483647 , +2147483647).

For example, the current absolute pulse count can be any value.

Send FA 01 FE 02 58 02 00 00 0C 80 CRC

The motor runs at a speed of 600 RPM and an acceleration of 2, reaching absolute 0x0C80 .

That is, after the motor finishes running, the current absolute pulse count becomes 0x0C80 .

For example, the current absolute pulse count can be any value.

Send FA 01 FE 02 58 02 FF FF F3 80 CRC

The motor runs at a speed of 600 RPM, an acceleration of 2, and an absolute value of -0x0C80 .

That is, after the motor finishes running, the current absolute pulse count becomes -0x0C80 .



Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	FEH	status (uint8_t)	CRC

status = 0 Start failed

status = 1 Start running

status = 2 Position completed

status = 3 Stopped upon touching limit switch

status = 5 Synchronous mode has received the instruction (it will only start running after receiving the synchronization flag).

Note: You can set whether to return to the running state using the command " 8CH ".



10.2.2 Stop command

The stop command is as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Bytes 4-5	Byte6	Bytes7-10	Bytes11
Header	Address	Code	data	acceleration	data	Checksum
FA	addr	FEH	00	acc	00	CRC

The stop command can control the motor to slow down and stop gradually, or it can control the motor to stop immediately. When acc is set to $\neq 0$, the motor decelerates and stops slowly. When acc is set to $= 0$, the motor stops immediately.

① Slow down and stop command (acc $\neq 0$)

for example:

Send FA 01 FE 00 00 02 00 00 00 00 FB
stop rotating at a deceleration of acc=2.

② Immediate stop command (acc = 0)

for example:

Send FA 01 FE 00 00 00 00 00 00 00 F9
Stop the motor immediately

Note: It is not recommended to use the immediate stop command when the motor speed exceeds 1000 RPM !

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	FEH	status (uint8_t)	CRC

Status = 0 Stop failed
status = 1 Stop Start
status = 2 position stopped completing
status = 3 Stopped upon touching limit switch

Note: You can also use the emergency stop command F7H to stop the operation.



10.2.3 Example of Pulse Absolute Operation

When operating in absolute pulse mode, the "zero point" pulse position needs to be set first. There are two ways to set it:

1. "Return to Zero" setting;
2. Setting the "92H command mode";

If not set, the default power-on location is "midnight".

1. **Set working mode**

FA 01 82 05 82

2. **Save settings**

FA 01 60 01 5C

3. **Set zero point**

FA 01 92 00 8D

4. **Set the running speed and absolute pulse count.**

FA 01 FE 01 2C 02 00 01 00 00 29

(Run at speed = 300 RPM , acc = 2, to pulse position 0x10000)

5. **After the process is complete, you can view the number of pulses.**

FA 01 33 2E

6. **If you need to slow down and stop the machine during operation, the command is as follows:**

FA 01 FE 00 00 02 00 00 00 00 FB

7. **If an emergency stop is required during operation, the command is as follows:**

FA 01 F7 F2

```
[2025-10-15 17:04:45.667]# SEND HEX>
FA 01 82 05 82
[2025-10-15 17:04:45.733]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 17:04:50.299]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 17:04:50.348]# RECV HEX>
FB 01 60 01 5D
[2025-10-15 17:04:57.932]# SEND HEX>
FA 01 92 00 8D
[2025-10-15 17:04:57.992]# RECV HEX>
FB 01 92 01 8F
[2025-10-15 17:05:04.579]# SEND HEX>
FA 01 FE 01 2C 02 00 01 00 00 29
[2025-10-15 17:05:04.647]# RECV HEX>
FB 01 FE 01 FB
[2025-10-15 17:05:12.122]# RECV HEX>
FB 01 FE 02 FC
[2025-10-15 17:05:18.579]# SEND HEX>
FA 01 33 2E
[2025-10-15 17:05:18.643]# RECV HEX>
FB 01 33 00 01 00 00 30
```



10.3 Relative motion according to coordinate values

In this mode, the motor can be controlled to move to a specified position relative to the coordinate values at a set acceleration and speed.

Note: The coordinate values are the cumulative multi-turn encoder values (16384/ turn), read using the command "31H".

10.3.1 Coordinate relative running instructions

The execution command format is as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Byte4-5	Byte6	Bytes7-10	Bytes11
Header	Addr	Code	speed	ACC	relative coordinates	Checksum
FA	addr	F4H	speed	acc	relAxis	CRC

The parameters are explained below:

speed speed value range 0 - 3000 (RPM)

acc accelerator value range 0 - 255.

relAxis the relative coordinate value, with a range of int32_t
(-2147483647 , +2147483647) .

For example, the current coordinate is 0x8000

Send FA 01 F4 02 58 02 00 00 40 00 8B

The motor operates at a speed of 600 RPM, an acceleration of 2, and a relative speed of 0x4000 .

That is, after the motor finishes running, the current coordinate becomes 0xC000 .

For example, the current coordinate is 0x8000

Send FA 01 F4 02 58 02 FF FF C0 00 09

The motor operates at a speed of 600 RPM, an acceleration of 2, and a relative speed of -0x4000 .

That is, after the motor finishes running, the current coordinate becomes 0x4000 .

Note: The "31H" command can read coordinate values.



Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	F4H	status (uint8_t)	CRC

status = 0 start failed
status = 1 start running
status = 2 position completed
status = 3 Stopped upon touching limit switch
status = 5 Synchronous mode has received the instruction (it
will only start running after receiving the synchronization flag).



10.3.2 Stop command

The stop command is as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Bytes 4-5	Byte6	Bytes 7-10	Bytes11
Header	Address	Code	data	ACC	data	Checksum
FA	addr	F4H	00	acc	00	CRC

The stop command can control the motor to slow down and stop gradually, or it can control the motor to stop immediately. When acc is set to $\neq 0$, the motor decelerates and stops slowly. When acc is set to $= 0$, the motor stops immediately.

① Slow down and stop command (acc $\neq 0$)

for example:

Send FA 01 F4 00 00 02 00 00 00 00 F1
stop rotating at a deceleration of acc=2.

② Immediate stop command (acc = 0)

for example:

Send FA 01 F4 00 00 00 00 00 00 00 EF
Stop the motor immediately

Note: It is not recommended to use the immediate stop command when the motor speed exceeds 1000 RPM !

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	F4H	status (uint8_t)	CRC

Status = 0	Stop failed
status = 1	Stop Start
status = 2	position stopped completing
status = 3	Stopped upon touching limit switch

Note: You can also use the emergency stop command F7H to stop the operation.



10.3.3 Example of running with coordinate coordinates

1. Set working mode
FA 01 82 05 82
2. Save settings
FA 01 60 01 5C
3. Set the running direction, speed, and relative coordinates.
FA 01 F4 01 2C 02 00 02 80 00 A0
(with speed = 300 (RPM) , acc = 2, relative to coordinates 0x28000)
4. After the process is complete, you can view the coordinates.
FA 01 31 2C
5. If you need to slow down and stop the machine during operation, the command is as follows:
FA 01 F4 00 00 02 00 00 00 00 F1
6. If an emergency stop is required during operation, the command is as follows:
FA 01 F7 F2

```
[2025-10-15 17:13:01.867]# SEND HEX>
FA 01 82 05 82
[2025-10-15 17:13:01.931]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 17:13:06.396]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 17:13:06.459]# RECV HEX>
FB 01 60 01 5D
[2025-10-15 17:13:11.556]# SEND HEX>
FA 01 F4 01 2C 02 00 02 80 00 A0
[2025-10-15 17:13:11.634]# RECV HEX>
FB 01 F4 01 F1
[2025-10-15 17:13:16.793]# RECV HEX>
FB 01 F4 02 F2
[2025-10-15 17:13:18.275]# SEND HEX>
FA 01 31 2C
[2025-10-15 17:13:18.337]# RECV HEX>
FB 01 31 00 00 00 02 7F FF AD |
```



10.4 Absolute motion based on coordinate values

In this mode, the motor can be controlled to run to the specified position with a set acceleration and speed, based on the coordinate values.

Note 1: The coordinate values are the cumulative multi-turn encoder values (16384/ turn), read using the command "3 1".

Note 2 : Supports real-time updates of speed and coordinates, meaning that a new command can be issued to change the speed and coordinates while the previous command is running.

Note 3: Before using absolute motion, a zeroing process must be performed once (instruction 0x9 1 or 0x9 2) to mark the "zero point".

10.4.1 absolute coordinate execution command

The execution command format is as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Byte4-5	Byte6	Bytes 7-10	Bytes11
Header	Address	Code	speed	acceleration	absolute coordinates	Checksum
FA	addr	F5H	speed	acc	absA xis	CRC

The parameters are explained below:

speed Speed value range 0 – 3000 (RPM)

acc Accelerator value range 0 – 255.

absAxis is the absolute coordinate value, with a range of int32_t (-2147483647 , +2147483647) .

For example, the current coordinates are any value

Send FA 01 F5 02 58 02 00 00 40 00 CRC

The motor runs at a speed of 600 RPM, with an acceleration of 2, and an absolute speed of 0x4000 .

That is, after the motor finishes running, the current coordinate becomes 0x4000 .

For example, the current coordinates are any value

Send FA 01 F5 02 58 02 FF FF C0 00 CRC

The motor runs at a speed of 600 RPM, with an acceleration of 2, and an absolute speed of -0x4000 .

That is, after the motor finishes running, the current coordinate becomes -0x4000 .



Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	F5H	status (uint8_t)	CRC

status = 0 start failed

status = 1 start running

status = 2 position completed

status = 3 Stopped upon touching limit switch

status = 5 Synchronous mode has received the instruction (it will only start running after receiving the synchronization flag).



10.4.2 Stop command

The stop command is as follows:

Downlink frames (PC → SERV0xxES)						
Byte1	Byte2	Byte3	Bytes4-5	Byte6	Bytes7-10	Bytes11
Header	Address	Code	data	acceleration	data	Checksum
FA	addr	F5H	00	acc	00	CRC

The stop command can control the motor to slow down and stop gradually, or it can control the motor to stop immediately. When acc is set to $\neq 0$, the motor decelerates and stops slowly. When acc is set to $= 0$, the motor stops immediately.

① Slow down and stop command (acc $\neq 0$)

for example:

Send FA 01 F5 00 00 02 00 00 00 00 F2
stop rotating at a deceleration of acc=2.

② Immediate stop command (acc = 0)

for example:

Send FA 01 F5 00 00 00 00 00 00 00 F0
Stop the motor immediately

Note: It is not recommended to use the immediate stop command when the motor speed exceeds 1000 RPM !

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	F5H	status (uint8_t)	CRC

Status = 0	Stop failed
status = 1	Stop/ Start
status = 2	position stopped completing
status = 3	Stopped upon touching limit switch

Note: You can also use the emergency stop command F7H to stop the operation.



10.4.3 Example of running with absolute coordinates

You need to first set the "zero point" coordinates. There are two ways to do this:

1. "Return to Zero" setting;
2. Setting the "92H command mode";

If not set, the default power-on position is the "zero point" coordinate.

1. **Set working mode**

FA 01 82 05 82

2. **Save settings**

FA 01 60 01 5C

3. **Set zero point**

FA 01 92 00 8D

4. **Set running speed, absolute coordinates**

FA 01 F5 01 2C 02 00 02 80 00 A1

(Run at speed = 300RPM, acc = 2, to pulse position 0x28000)

5. **After the process is complete, you can view the coordinates.**

FA 01 31 2C

6. **If you need to slow down and stop the machine during operation, the command is as follows:**

FA 01 F5 00 00 02 00 00 00 00 F2

7. **If an emergency stop is required during operation, the command is as follows:**

FA 01 F7 F2

```
[2025-10-15 17:20:24.915]# SEND HEX>
FA 01 82 05 82
[2025-10-15 17:20:24.967]# RECV HEX>
FB 01 82 01 7F
[2025-10-15 17:20:29.347]# SEND HEX>
FA 01 60 01 5C
[2025-10-15 17:20:29.423]# RECV HEX>
FB 01 60 01 5D
[2025-10-15 17:20:34.051]# SEND HEX>
FA 01 92 00 8D
[2025-10-15 17:20:34.118]# RECV HEX>
FB 01 92 01 8F
[2025-10-15 17:20:38.868]# SEND HEX>
FA 01 F5 01 2C 02 00 02 80 00 A1
[2025-10-15 17:20:38.929]# RECV HEX>
FB 01 F5 01 F2
[2025-10-15 17:20:44.146]# RECV HEX>
FB 01 F5 02 F3
[2025-10-15 17:20:45.427]# SEND HEX>
FA 01 31 2C
[2025-10-15 17:20:45.493]# RECV HEX>
FB 01 31 00 00 00 02 7F FB A9
```



10.4.4 Example of real-time updates with absolute coordinates

This mode supports real-time updates of speed and coordinates.

You need to first set the "zero point" coordinates. There are two ways to do this:

1. "Return to Zero" setting;
2. Setting the "92H command mode";

If not set, the default power-on position is the "zero point" coordinate.

1. **Set working mode**

FA 01 82 05 82

2. **Save settings**

FA 01 60 01 5C

3. **Set zero point**

FA 01 92 00 8D

4. **Set running speed, absolute coordinates**

FA 01 F5 01 2C 02 70 A2 80 00 B1

(Run at speed = 300RPM, acc = 2, to pulse position 0x70A28000)

Waiting about 20 seconds for the motor to run, then update the speed and coordinate values.

5. **Update the running speed, absolute coordinates**

FA 01 F5 02 58 02 00 02 80 00 CE

(Run at speed = 600RPM, acc = 2, to pulse position 0x28000)

The motor will change speed and target position during operation.

6. **After the process is complete, you can view the coordinates.**

FA 01 31 2C

```
[2025-11-21 14:48:21.613]# SEND HEX>
FA 01 82 05 82
[2025-11-21 14:48:21.682]# RECV HEX>
FB 01 82 01 7F
[2025-11-21 14:48:26.813]# SEND HEX>
FA 01 60 01 5C
[2025-11-21 14:48:26.872]# RECV HEX>
FB 01 60 01 5D
[2025-11-21 14:48:32.023]# SEND HEX>
FA 01 92 00 8D
[2025-11-21 14:48:32.086]# RECV HEX>
FB 01 92 01 8F
[2025-11-21 14:48:47.533]# SEND HEX>
FA 01 F5 01 2C 02 70 A2 80 00 B1
[2025-11-21 14:48:47.602]# RECV HEX>
FB 01 F5 01 F2
[2025-11-21 14:49:07.957]# SEND HEX>
FA 01 F5 02 58 02 00 02 80 00 CE
[2025-11-21 14:49:08.019]# RECV HEX>
FB 01 F5 01 F2
[2025-11-21 14:49:27.961]# RECV HEX>
FB 01 F5 02 F3
[2025-11-21 14:49:29.605]# SEND HEX>
FA 01 31 2C
[2025-11-21 14:49:29.674]# RECV HEX>
FB 01 31 00 00 00 02 80 01 B0
```

Part 11. Multi- motor synchronous motion description

Synchronous motion refers to starting multiple motors connected to a single bus to move simultaneously. There are 4 ways to achieve this:

Method 1:

Sending a command using broadcast address 0 will cause all motors on the bus to execute the command, which is suitable for all motors on the bus to execute the same command.

Method 2:

When a command is sent using a group address, all motors in the same group on the bus will execute it. This is suitable for group execution of commands by motors on the bus.

Method 3:

Using multi-instruction data frames, a single instruction contains different control commands for multiple motors, suitable for different motors on the bus to execute different motion commands. This method can control up to 5 motors simultaneously.

Method 4:

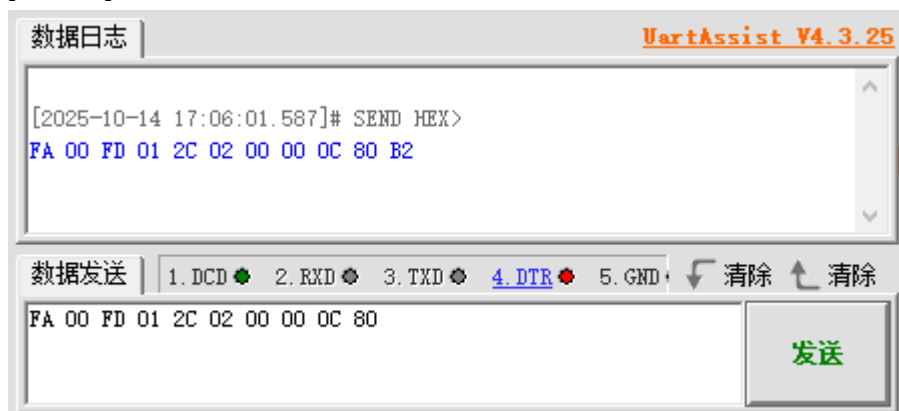
The multi-motor synchronization flag function is used to enable different motors on the bus to execute different motion commands. This method allows for the control of an unlimited number of motors.

11.1 synchronous motion mode 1 - Broadcast

To send motor movement commands using broadcast address 0, such as instructing all motors on the bus to move relative to each other for 3200 pulses, the following command can be sent.

FA 00 FD 01 2C 02 00 00 0C 80 B2

All motors on the bus will move relative to each other for 3200 pulse positions.





11.2 synchronous motion mode 2 - grouping method

To send motor movement commands using group addresses, such as instructing all motors with group address 0x50 on the bus to run relative to each other for 3200 pulses, the following command can be sent.

FA 50 FD 01 2C 02 00 00 0C 80 02

All motors with group address 0x50 on the bus will run for 3200 pulse positions.

See the section 4.3.12 for group address settings .

11.3 synchronous motion mode 3 - Multi-command data frame

Multi-command data frames, meaning a single frame can contain up to five commands. The slave determines which command to execute based on the address. This means a single frame can control up to five motors.

If there are more than five motors, multi-motor synchronous control can be used; see Section 6.9.

If there are fewer than five motors, all remaining command Bytes are set to 0.

Long data packets format:

Head	0xFC				1Byte
	Byte 1	Byte 2	...	Byte 10	
command 1	slaveAddr1	code	...		10 Byte
command 2	slaveAddr 2	code	...		10 Byte
command 3	slaveAddr 3	code	...		10 Byte
command 4	slaveAddr 4	code	...		10 Byte
command 5	slaveAddr 5	code	...		10 Byte
checksum	CRC				1 Byte

Note:

- 1.The length of the long data packet is 52 Bytes in total.
- 2.The length of each command X is Byte10, when it is less than Byte10, add 0 to supplement.
- 3.Command X is the corresponding ordinary command, remove the frame header (FA) and checksum.
- 4.If the slave addresses of command X and command Y (X<Y) are the same, only command X is executed.
- 5.Slave does not answer.



For example, sending the following long data packet can control 5 motors to perform different actions (16 subdivisions)

FC

01 F6 00 32 0A 00 00 00 00 00

02 FD 01 2C 02 00 04 E2 00 00

03 FE 02 58 02 00 04 E2 00 00

04 F4 02 58 64 00 0C 80 00 00

05 F5 04 B0 C8 00 0C 80 00 00

CA

FC 01 F6 00 32 0A 00 00 00 00 00 02 FD 01 2C 02 00 04 E2 00 00 03 FE 02
58 02 00 04 E2 00 00 04 F4 02 58 64 00 0C 80 00 00 05 F5 04 B0 C8 00 0C
80 00 00 CA

Motor 1 rotates continuously forward in speed mode (speed=0x32, acc=0x0A).

Motor 2 rotates forward 100 times in position mode 1 (speed=0x12C, acc=0x02).

Motor 3 rotates forward 100 times in position mode 2 (speed=0x258, acc=0x02).

Motor 4 moves to coordinate 0xC8000 in position mode 3 (speed=0x258, acc=0x64).

Motor 5 moves to coordinate 0xC8000 in position mode 4 (speed=0x4B0, acc=0xC8).



11.4 synchronous motion mode 4 - Synchronization mark

After using the synchronization flag function, the motor will not execute the motion command immediately after receiving it. It will wait for the synchronization execution command to be received before it can start executing.

This method allows control of all motors on the bus.

11.4.1 Command to enable/disable synchronization function

The configuration command is as follows:

Downlink frames (PC → SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FA	addr	4AH	enable	CRC

enable = 00 disables the synchronization function (default).

enable = 01 Enables the synchronization function.

Returned data:

Uplink frames (PC ← SERV0xxES)				
Byte1	Byte2	Byte3	Byte4	Byte5
Header	Address	Code	Setting status	Checksum
FB	addr	4AH	status (uint8_t)	CRC

status = 0 Set failed.

status = 1 set succeeded.

11.4.2 Synchronous execution instructions

The motor only begins to execute the previously received motion commands after receiving the synchronization execution command.

The synchronous execution instructions are as follows:

Downlink frames (PC → SERV0xxES)			
Byte1	Byte2	Byte3	Byte4
Header	Address	Code	Checksum
FA	00	4BH	CRC

Returned data: None.

If there are many motors on the bus, some motors may not receive the synchronization execution command due to interference or other reasons. In this case, consider sending the synchronization execution command repeatedly at intervals of about 1ms.



11.4.3 Example of multi-motor synchronous operation

Taking a bus with 5 motors as an example, different motion commands are executed synchronously.

1. Setting the operating mode: RS485 bus closed-loop FOC mode

Motor 1: FA 01 82 05 CRC

...

Motor 5: FA 01 82 05 CRC

2. Set 4AH parameters: Enable multi-motor synchronous control function.

Motor 1: FA 01 4A 01 CRC

...

Motor 5: FA 05 4A 01 CRC

3. Set the motion mode and parameters for each motor separately.

(1) Motor 1: Speed mode, forward rotation, speed 640 RPM, acceleration 2

FA 01 F6 0 2 80 02 75

(2) Motor 2: Position control mode 1, reverse, speed 100 RPM, acceleration 2, 320000 pulses (100 revolutions in 16 microsteps).

FA 0 2 FD 8 0 64 02 00 04 E2 00 CRC

(3) Motor 3: Position control mode 2, reverse, speed 100 RPM, acceleration 2, 320000 pulses (100 revolutions in 16 microsteps).

FA 03 FE 8 0 64 02 00 04 E2 00 CRC

(4) Motor 4: Position control mode 3, forward rotation, speed 300 RPM, acceleration 200, running to coordinate 0x280000 (160 revolutions in 16 microsteps).

FA 04 F4 0 1 2C C8 02 80 00 00 CRC

(5) Motor 5: Position control mode 4, forward rotation, speed 300 RPM, acceleration 200, running to coordinate 0x280000 (160 revolutions in 16 microsteps).

FA 05 F5 0 1 2C C8 02 80 00 00 CRC

4. Send synchronization command: All motors with the above-configured parameters will begin synchronous operation.

FA 00 4B CRC (1MS interval)

FA 00 4B CRC (1MS interval)

...

5. Motor stop: During operation, a command can be issued to stop the motor.

FA 01 F7 CRC

...

FA 05 F7 CRC



See the image below for detailed configuration instructions:

```
[2025-08-19 11:43:09.611]# SEND HEX>
FA 01 82 04 81
[2025-08-19 11:43:09.673]# RECV HEX>
FB 01 82 01 7F
[2025-08-19 11:43:15.139]# SEND HEX>
FA 02 82 04 82
[2025-08-19 11:43:15.200]# RECV HEX>
FB 02 82 01 80
[2025-08-19 11:43:19.596]# SEND HEX>
FA 03 82 04 83
[2025-08-19 11:43:19.656]# RECV HEX>
FB 03 82 01 81
[2025-08-19 11:43:23.332]# SEND HEX>
FA 04 82 04 84
[2025-08-19 11:43:23.394]# RECV HEX>
FB 04 82 01 82
[2025-08-19 11:43:27.771]# SEND HEX>
FA 05 82 04 85
[2025-08-19 11:43:27.831]# RECV HEX>
FB 05 82 01 83
[2025-08-19 11:43:36.508]# SEND HEX>
FA 01 4A 01 46
[2025-08-19 11:43:36.575]# RECV HEX>
FB 01 4A 01 47
[2025-08-19 11:43:40.652]# SEND HEX>
FA 02 4A 01 47
[2025-08-19 11:43:40.724]# RECV HEX>
FB 02 4A 01 48
[2025-08-19 11:43:42.757]# SEND HEX>
FA 03 4A 01 48
[2025-08-19 11:43:42.821]# RECV HEX>
FB 03 4A 01 49
[2025-08-19 11:43:44.197]# SEND HEX>
FA 04 4A 01 49
[2025-08-19 11:43:44.259]# RECV HEX>
FB 04 4A 01 4A
[2025-08-19 11:43:45.748]# SEND HEX>
FA 05 4A 01 4A
[2025-08-19 11:43:45.810]# RECV HEX>
FB 05 4A 01 4B

[2025-08-19 11:43:54.686]# SEND HEX>
FA 01 F6 02 80 02 75
[2025-08-19 11:43:54.746]# RECV HEX>
FB 01 F6 05 F7
[2025-08-19 11:44:01.765]# SEND HEX>
FA 02 FD 80 64 02 00 04 E2 00 C5
[2025-08-19 11:44:01.823]# RECV HEX>
FB 02 FD 05 FF
[2025-08-19 11:44:08.515]# SEND HEX>
FA 03 FE 80 64 02 00 04 E2 00 C7
[2025-08-19 11:44:08.570]# RECV HEX>
FB 03 FE 05 01
[2025-08-19 11:44:16.100]# SEND HEX>
FA 04 F4 01 2C C8 02 80 00 00 69
[2025-08-19 11:44:16.152]# RECV HEX>
FB 04 F4 05 F8
[2025-08-19 11:44:24.669]# SEND HEX>
FA 05 F5 01 2C C8 02 80 00 00 6B
[2025-08-19 11:44:24.736]# RECV HEX>
FB 05 F5 05 FA
[2025-08-19 11:44:45.572]# SEND HEX>
FA 00 4B 45
[2025-08-19 11:44:53.749]# SEND HEX>
FA 01 F7 F2
[2025-08-19 11:44:53.809]# RECV HEX>
FB 01 F7 01 F4
[2025-08-19 11:44:57.059]# SEND HEX>
FA 02 F7 F3
[2025-08-19 11:44:57.119]# RECV HEX>
FB 02 F7 01 F5
[2025-08-19 11:44:59.643]# SEND HEX>
FA 03 F7 F4
[2025-08-19 11:44:59.700]# RECV HEX>
FB 03 F7 01 F6
[2025-08-19 11:45:02.075]# SEND HEX>
FA 04 F7 F5
[2025-08-19 11:45:02.129]# RECV HEX>
FB 04 F7 01 F7
[2025-08-19 11:45:04.932]# SEND HEX>
FA 05 F7 F6
[2025-08-19 11:45:04.989]# RECV HEX>
FB 05 F7 01 F8
```

Part 12. Serial Port Assistant Setup Instructions

12.1 Serial Port Assistant Configuration Example

Select the serial port number: (COMxx)

Select baud rate: 38400

Select check digit : NONE

Select data bits : 8

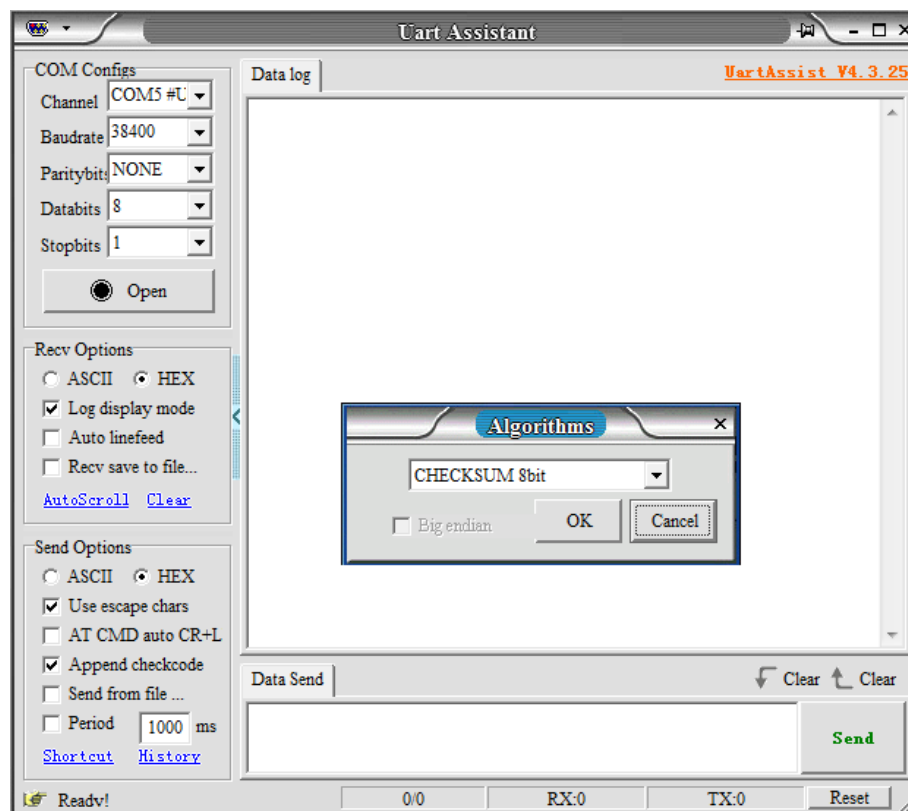
Select stop bit: 1

Receive settings, select: Hex

Send settings, select: Hex

Automatically send checksum bit , select: CHECKSUM 8 bit

As shown in the image below:

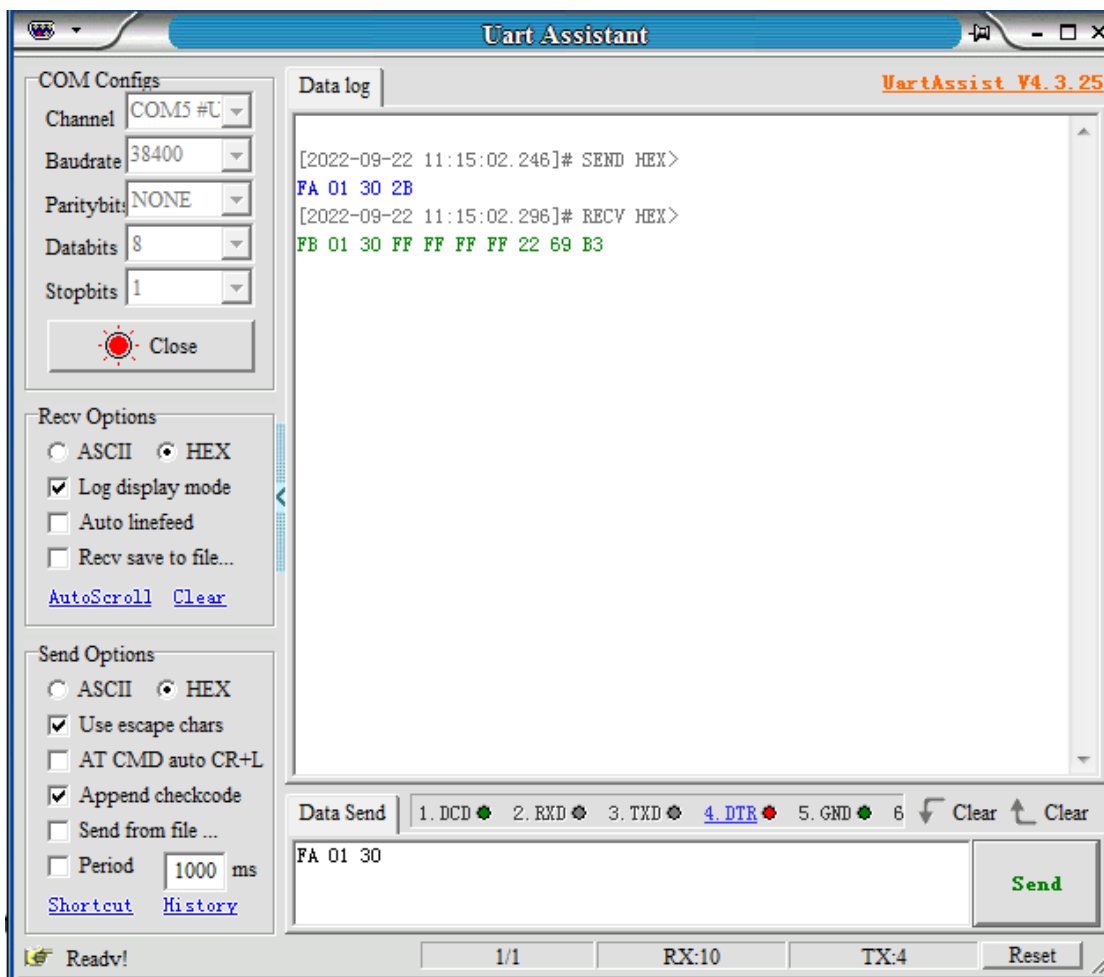




12.2 Example of reading encoder values

Send **FA 01 30 2B**

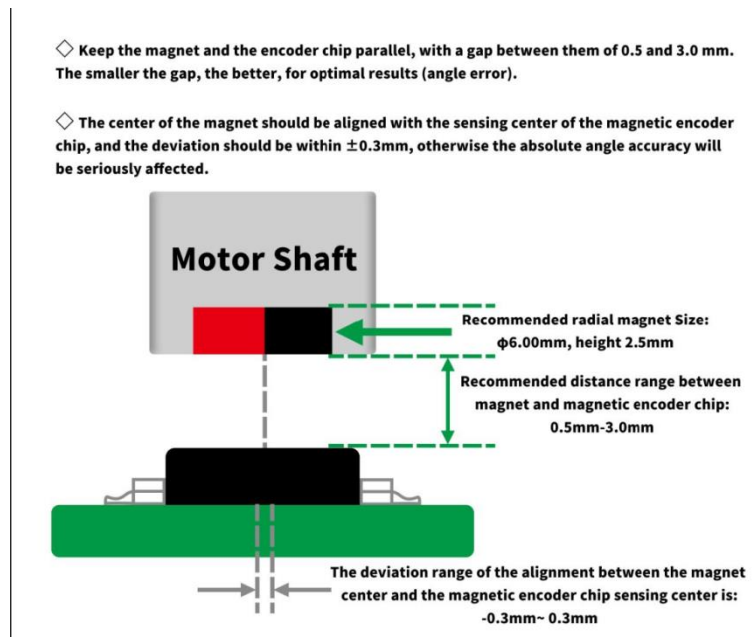
Return to **FB 01 30 FF FF FF FF 22 69 B3**



Part 13. Frequently Asked Questions and Precautions

13.1 Precautions

1. Power input voltage: 20V - 48V ;
2. Do not plug or unplug the power cord or signal cable while the circuit is powered on, as this may damage the driver board .
3. The A+, A-, B+, and B- phase lines must correspond; otherwise, an error will occur.
4. We recommend that users purchase our matching motors directly to avoid the risk of incompatibility. If you choose to use a custom motor in closed-loop mode instead of our motors, the following conditions must be met:
 - (1) The motor step distance is 1.8 degrees.
 - (2) The motor's internal resistance is less than 10 ohms.
 - (3) A radial magnet can be installed on the back of the motor.
 - (4) When installing magnets, please note the following diagram:



5. After the magnets are installed, a calibration procedure must be performed to compensate for any installation deviations of the magnets; refer to section 4.1.



13.2 Frequently Asked Questions

Serial Number	question	Solution
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
1 2		

Part 14. contact us

<https://makerbase.aliexpress.com/>

<https://www.youtube.com/channel/UC2i5I1tcOXRJ2ZJiRxwpCUQ>

<https://github.com/makerbase-motor>